Male Earnings Volatility in LEHD before, during, and after the Great Recession

by

Kevin L. McKinney
U.S. Census Bureau

John M. Abowd
U.S. Census Bureau

CES 20-31 **September, 2020**

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Abstract

This paper is part of a coordinated collection of papers on prime-age male earnings volatility. Each paper produces a similar set of statistics for the same reference population using a different primary data source. Our primary data source is the Census Bureau's Longitudinal Employer Household Dynamics (LEHD) infrastructure files. Using LEHD data from 1998 to 2016, we create a well-defined population frame to facilitate accurate estimation of temporal changes comparable to designed longitudinal samples of people. We show that earnings volatility, excluding increases during recessions, has declined over the analysis period, a finding robust to various sensitivity analyses. Although we find volatility is declining, the effect is not homogeneous, particularly for workers with tenuous labor force attachment for whom volatility is increasing. These "not stable" workers have earnings volatility approximately 30 times larger than stable workers, but more important for earnings volatility trends we observe a large increase in the share of stable employment from 60% in 1998 to 67% in 2016, which we show to largely be responsible for the decline in overall earnings volatility. To further emphasize the importance of not stable and/or low earning workers we also conduct comparisons with the PSID and show how changes over time in the share of workers at the bottom tail of the cross-sectional earnings distributions can produce either declining or increasing earnings volatility trends.

[*] Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the US Census Bureau or other sponsors. All results have been reviewed to ensure that no confidential information is disclosed (DRB clearance numbers DRB-B0066-CED-20190712, CBDRB-FY20-279, and CBDRB-FY20-288). This research uses data from the US Census Bureau's Longitudinal Employer-Household Dynamics Program, which was partially supported by NSF grants SES-9978093, SES-0339191, and ITR-0427889; National Institute on Aging grant AG018854; and grants from the Alfred P. Sloan Foundation. The most recent version of this paper can be found at http://arxiv.org/abs/2008.00253.

I. Introduction

This paper is part of a coordinated collection of papers on prime-age male earnings volatility. Each paper produces a similar set of statistics for the same reference population using a different primary data source. Our paper uses data from the Census Bureau's Longitudinal Employer Household Dynamics (LEHD) program. The other three papers are Carr et al. (2020), who use the Survey of Income and Program Participation (SIPP) linked to administrative data, Moffit and Zhang (2020), who use the Panel Study of Income Dynamics (PSID), and Zilliak et al. (2020), who use the Current Population Survey Annual Social and Economic Supplement (CPS-ASEC) linked to administrative data. Comparing the results in all four papers, reveals that each data source has both advantages and shortcomings. The authors hope, collectively, that by using a variety of primary data sources we can take advantage of each source's strengths while judiciously combining data sources to assess each one's weaknesses, thus forming a more complete picture of the changes in earnings volatility over the past twenty years.

To put our paper in context, we note it is based on methods developed in Abowd et al. (2018; AMZ hereafter). That paper, too, was part of a collection of related papers on earnings inequality. Among other findings, AMZ demonstrated that in order to draw comparisons between administrative, designed cross-sectional, and designed longitudinal estimates of related labor market phenomena like earnings inequality and volatility, one must construct a proper, dynamic universe as a reference population. Armed with that construct, which we also employ in this paper, AMZ show, first, that careful attention to the left tail (bottom) of the earnings distribution is essential to understanding both the cross-sectional and longitudinal patterns in earnings and earnings volatility. Second, AMZ demonstrate that careful attention to individuals with zero earnings for a full calendar year is also critical to understanding earnings outcomes, particularly in dynamic settings. Both of these conclusions figure prominently in the statistical analysis we present in this paper.

Using LEHD data prepared using our consistent, dynamic population frame, we estimate earnings volatility trends for prime-age males from 1998 to 2016. In contrast with survey data sources such as the CPS, PSID, and ACS, LEHD data contain annual earnings for the virtual universe of private wage and salary workers in the United States. The essentially complete coverage of the universe enables a detailed analysis of worker earnings volatility. However, as noted above, it is essential to use earnings associated with only "eligible" workers—those who

are part of our dynamic frame—namely, individuals with identifiers issued by the Social Security Administration, and used by the person to whom they were issued. Adopting this population frame, allows us to consistently estimate calendar-year time trends for our earnings volatility measures.

Unlike many other studies, we estimate earnings volatility both with and without years with zero earnings. This is potentially important, especially post Great Recession, when large numbers of workers were forced to transition to inactive status and thus had zero earnings, often for multiple calendar years. One caveat with including years with zero earnings is that unlike survey data, LEHD data contains no affirmative report of zero earnings, the zero is assumed based on the absence of reported earnings, which is only possible because the analysis sample is based on our independently constructed consistent, dynamic universe. To minimize earnings under-reporting, we also impute earnings for workers at firms suspected of under/non-reporting.

Moffitt and Zhang (2018a and 2018b) have two excellent earnings volatility literature reviews, which we briefly touch on here. Many past studies on earnings inequality and volatility use variations or extensions of the estimation framework pioneered in Gottschalk and Moffitt (1994). These studies estimate an error components model with both an annual (biennial for the PSID) permanent and transitory component. Assuming the arguably strong assumptions embedded in the Gottschalk-Moffitt model are correct, estimates of average annual earnings volatility can be recovered for both components.

We take a less structured approach and do not attempt to decompose earnings into a permanent and transitory component. Instead, we estimate two different measures of gross earnings volatility (Difference in Log Earnings and Arc Percentage Change). Our main result is volatility has declined over the analysis period, excluding increases during recessions, a finding robust to the exclusion of outliers (trimming) and the inclusion of workers with zero earnings (entrants and exiters). This result is not surprising and is consistent with previous research using administrative data showing declining or no trend in earnings volatility (Sabelhaus and Song 2009, 2010; Celik et al. 2012; and Guvvenen, Ozkan, and Song 2014).

Although volatility in the LEHD data is declining overall, there are two distinct periods; the period prior to the great recession where there is no trend and the period after the great recession where there is a declining trend. For the typical worker the benefits of the decrease in earnings volatility post great recession are mixed. On the one hand, the share of workers whose

earnings increase is relatively high, however the increases are smaller on average and the variance is low (potentially limiting upward mobility).

Recent research by Jensen and Shore (2015) and Arellano, Blundell, and Bonhomme (2017) highlight how different data generating processes can create similar changes in average earnings volatility. They find earnings volatility is not homogenous, the typical worker has relatively low earnings volatility and changes in average volatility are driven by a smaller subset of workers with repeated large earnings shocks. These results are consistent with our prior research and the results in this paper showing that most workers typically do not have large year-to-year earnings changes. Assuming there are two distinct types of employees, we classify workers every year as either stable (employed each quarter with no change in the dominant or highest earning employer) or not stable. Stable workers make up the largest share (65%) of all workers in 2008 and have low volatility combined with regular earnings increases. In contrast not stable workers have almost 30 times higher earnings volatility and are much more likely to have earnings decreases, especially during recessions. Of more relevance to the overall decline in earnings volatility is the relatively large shift of workers into stable employment, from 60% in 1998 to 67% in 2016. If this shift did not occur, ceteris paribus, the trend in overall earnings inequality would have increased rather than decreased as observed.

To further emphasize the importance of not stable and/or low earning workers we also conduct comparisons with the PSID and show how the composition of the bottom tail of the cross-sectional earnings distribution can produce either declining or increasing trends in earnings volatility. The lower tail of the PSID earning distribution is generally small relative to the lower tail in the LEHD earnings distribution, but the share of low earning workers is increasing in the PSID over the analysis period, while the share is relatively stable in LEHD data. Given the dominant role low earning workers play in the estimation of earnings volatility, we show how the expanding lower tail in the PSID is largely responsible for the positive trend in PSID earnings volatility. In contrast, the relatively stable lower tail of the earnings distribution found in the LEHD data clearly demonstrates the advantage of large-scale administrative data. LEHD data consistently captures low earning high volatility workers, enabling us to produce a robust estimate of declining earnings volatility.

Our paper is organized as follows. Section II describes the methods we used to construct the LEHD analysis samples. Section III presents our statistical results. Section IV concludes.

II. Data

The empirical work in this paper uses earnings information from the Longitudinal Employer-Household Dynamics (LEHD) infrastructure files, developed and maintained by the U.S. Census Bureau.[1] From this data source, we construct annual person-level earnings files covering the period 1998-2017.

In the LEHD data infrastructure, a "job" is the statutory employment of a worker by a statutory employer as defined by the Unemployment Insurance (UI) system in a given state. Mandated reporting of UI-covered wage and salary payments between one statutory employer and one statutory employee is governed by the state's UI system. Reporting covers private employers and state and local government. There are no self-employment earnings unless the proprietor drew a salary, which, for UI earnings data, is indistinguishable from other employees.

States joined the Local Employment Dynamics federal/state partnership that supplies input data to the LEHD program at different dates. When a state joined, the data custodians were asked to produce historical data for as many quarters in the past, back to 1990Q1, as could be reasonably recovered from their information storage systems. As a result, the date that a data-supplying entity joined the partnership is not the same as the first quarter in which that entity's data appear in the system. The start date for any state depends primarily on the amount of historical data the state could recover at the time it joined. This potential ignorability (in the sense of Rubin 1987 or Imbens and Rubin 2015) of the start date for a segment of the LEHD data – that is the possibility that state start dates are conceivably not related to data quality or earnings volatility – is the basis for our methods of constructing a time-series of nationally representative estimates.

Although state entry is a potential concern, AMZ show that the annual earnings distributions for the subset of states available by 1995Q1 are almost identical to the complete data annual earnings distribution. However, in this paper we are not estimating measures of earnings inequality but are instead measuring earnings volatility, primarily using the variance of the difference in log annual real earnings. The variance of the difference in log annual earnings is especially sensitive to earnings changes at the bottom of the earnings distribution, where a

[1] See Abowd et al. (2009) for a detailed summary of the construction of the LEHD infrastructure.

relatively large change in absolute value can result in a large percentage change. Even though AMZ show the annual earnings distributions between the fifth and the ninety-fifth percentiles are not noticeably affected by state entry/exit, we need to be cautious in applying their results to this paper because earnings changes for workers in the bottom five percent of the earnings distribution potentially have an outsized impact on the results.

Table 1 shows information about the available data for each of the fifty states, plus the District of Columbia.[2] By 1995Q1 25 states, including the larger states with relatively high shares of high earning workers (Illinois, California, Florida, New York, and Texas), are available in the LEHD infrastructure file system, representing about 68% of 2012 BLS Quarterly Census of Employment and Wages (QCEW) Month 1 employment. Between 1995Q2 and 1998Q1 fourteen new states enter the LEHD infrastructure file system, representing an additional 21% of 2012 BLS QCEW Month 1 employment. Thus, by 1998Q1 the LEHD infrastructure consists of 39 states representing approximately 89% of 2012 BLS QCEW month 1 employment. Given the relatively large share of employment entering the LEHD infrastructure (21%) between 1995Q1 and 1998Q1 and the sensitivity of the variance of the change in earnings to the composition of the left tail of the earnings distribution we have chosen to take a conservative approach and start our analysis in 1998Q1.

State entry continues over the ensuing years until 2003Q3 when the final state, Mississippi, is available for analysis. The complete data for all fifty-one states are available through 2015Q4, however in 2016Q1 Missouri exits the LEHD infrastructure file system. Alaska and South Dakota also exit prior to the end of the sample period in 2016Q3 and 2017Q2, respectively. To capture the different epochs of data availability, we create three sample regimes. Each sample regime consists of a set of states available each quarter over a pre-defined time period. The three sets of states and the time periods for each regime are chosen such that state entry/exit are forced to occur at two discrete points in time, thus allowing for a clear presentation of the cumulative effect of state entry/exit. The three sample regimes are defined in Table 1. The first sample regime consists of 39 states with data available beginning no later than 1998Q1 and ending no earlier than 2015Q4. The second sample regime consists of all fifty-one states beginning in 2004Q1 and ending in 2015Q4. The third sample regime consists of 48 states beginning in 2004Q1 and ending in 2017Q4.

[2] Going forward we will consider the District of Columbia to be equivalent to a state.

Although the LEHD data provide a high-quality jobs frame, individual identifier misuse complicates the time-varying many-to-one assignment of jobs to workers. Therefore, when studying earnings volatility, it is preferable to have a person frame that covers a known population of interest, such as all persons legally eligible to work in the United States. For our analysis, we create a frame of workers using the Census Bureau's edited version of the Social Security Administration's master SSN database (the "Census Numident"), capturing all officially reported employment-eligible workers but removing jobs associated with ineligible workers, as we elaborate below.

LEHD earnings records are reported quarterly by the employing firm. These records contain a nine-digit person identifier, typically assumed to be a Social Security Number. However, at the time the report is received by the state UI office, the nine-digit person identifier is not verified, resulting in records both with and without a valid SSN. Using the Census Numident we ascertain if each earnings record is associated with a valid SSN. Records not associated with a valid SSN may have an alternate valid person identifier such as an IRS-issued Individual Taxpayer Identification Number (ITIN); nevertheless, we can only distinguish between valid and invalid SSNs. If the SSN is valid, in addition to the UI employment history we have access to demographic characteristics, such as sex and birthdate, from the Census Numident.

Using both the Census Numident and the employment histories from the UI data, we create a "prime-age male eligible-workers" frame, including only workers in a given year that meet the following criteria:

- individual has a valid SSN on the Census Numident;
- gender of the individual is male;
- the year is between 1998 and 2017, inclusive;
- the modal age of the individual during the year is between 25 and 59, inclusive;
- the year is greater than the SSN year-of-issue and less than the year of death (if available); and
- the SSN is associated with fewer than 12 jobs during the year.

An eligible worker is labeled as "active" in the labor market when UI earnings are positive and "inactive" otherwise. Inactive status is thus inferred based on the absence of positive earnings reports.

The purpose of the prime-age male eligible-workers frame is twofold. First, the Census Numident data allow us to consistently identify a set of males legally eligible to work each year, while at the same time implicitly removing earning records from our analysis sample not associated with individuals in the covered legally eligible to work population. However, we go a step further, removing earnings records with valid SSNs where the available data strongly suggest the SSN is not being used by the person to whom it was issued.[3] Our criteria for excluding earnings records uses three annual exclusion rules, preserving job level earnings records during years when none of the rules are violated. The three exclusion rules are: job years with positive UI earnings prior to the SSN being issued are removed, while later years are eligible for inclusion; job years with positive reported earnings where the individual is reported dead prior to the start of the year are removed, although earlier years are eligible for inclusion; and job years where the individual has more than 12 employers are excluded, although other years are eligible for inclusion.

When studying earnings volatility, we are primarily concerned with calendar year to calendar year changes in earnings. To facilitate this, we transform the annual earnings data (1998-2017) into a year-pair dataset (1998-2016) where each observation contains information from the initial and the subsequent year. Table 2 shows counts (rounded to four significant digits) of our analysis sample of active, prime-age, male, eligible-workers by year-pair, broken down by labor market status (active both years or active in only one year) and sample regime. For some of our results we analyze the sample regimes separately, however we usually use a composite sample constructed from the three sample regimes. The composite sample is constructed each year by choosing the available sample regime with the largest number of states. Applying this rule results in a composite sample consisting of regime 1 from 1998-1999 to 2003-2004, regime 2 from 2004-2005 to 2014-2015, and regime 3 from 2015-2016 to 2016-2017.

Our earnings measure is based on annual UI job-level earnings reports. First, we adjust nominal earnings to real earnings using the Bureau of Economic Analysis' Personal Consumption Expenditures (PCE) index, with 2010 as the base year. Let v_{ijt} be the real earnings for worker i employed at firm j in year t. Second, we calculate person-level real annual earnings

[3] The use of SSNs not originally issued to the person using the SSN has been documented and studied by Brown et al 2013, AMZ and others.

e_{it} for each eligible male worker i in each year t as the sum of earnings at all jobs for each worker i in each year t[4]

$$e_{it} = \sum_{j} v_{ijt} \tag{1}$$

To estimate earnings volatility, we create two measures of the change in annual earnings. Our primary earnings volatility measure is the difference in log earnings from the initial year t to the subsequent year $t + 1$

$$l_{it} = \ln(e_{it+1}) - \ln(e_{it}) . \tag{2}$$

The difference in log earnings measure, l_{it}, is available from 1998 to 2016 for workers with positive earnings in both years. We also analyze a second measure, the arc percentage change a_{it} (Dahl, DeLeire, and Schwabish, 2008, 2011; Ziliak, Hardy, and Bollinger, 2011)

$$a_{it} = \frac{(e_{it+1} - e_{it})}{(e_{it} + e_{it+1})/2}. \tag{3}$$

The arc percentage change is available from 1998 to 2016 for all workers in the composite sample.

If the change in earnings is moderate, l_{it} and a_{it} produce similar results. For example, when earnings decrease by less than fifty percent or increase less than one hundred percent, the relative difference between the two measures is no more than three percentage points. However, when there is an extremely large percentage change in earnings the two measures may differ substantially. For example, if a worker earns $1,000 in the first year and $50,000 in the second year of a year-pair, the difference in log earnings is over twice as large as the arc percentage

[4] Although not discussed here, in section III.c. we also estimate volatility using age-adjusted real annual earnings.

change (3.91 vs 1.92). Given the sensitivity of the variance calculation to large earnings changes, the two measures may produce very different results even when presented with the same data.

Unlike the difference in log earnings measure l_{it}, the arc percentage change can be used when one of the earnings observations for the year-pair is zero. The arc percentage change is minus two when a worker has positive earnings in the first year and zero earnings in the second year (exiter) and is two when earnings in the first year are zero and earnings in the second year are positive (entrant). Otherwise, the arc percentage change is bounded in the open interval $(-2,2)$. The addition of entrants and exiters to the arc percentage change earnings volatility calculation unambiguously increases earnings volatility, providing a measure of the impact of long duration negative labor supply shocks.

III. Results

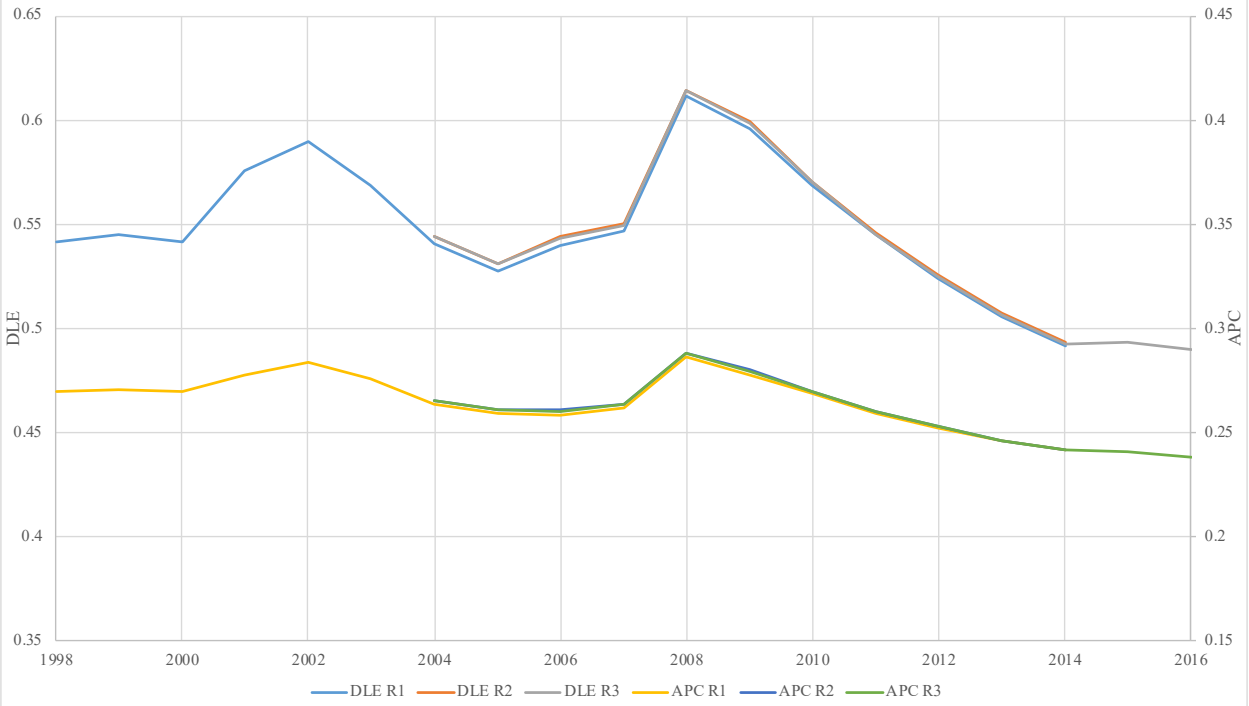
a. Sample Regimes

As a first check before formally presenting our results, we assess the impact of two alternating periods of state level missing data. At the beginning of our analysis period, twelve states representing approximately 11% of QCEW 2012 month 1 employment are missing, followed by a complete data period, while near the end of our sample three states that make up about 2.5% of employment are missing. If states are missing completely at random, then the level and trend of earnings volatility should not noticeably be affected during periods where states are missing relative to periods where the data are complete. In Figure 1, we show two different measures of earnings volatility: the first is the difference in log earnings (DLE), l_{it}, and the second is the arc percentage change (APC), a_{it}. For both measures, the calculation includes only workers with earnings in both the initial and the subsequent year. Three separate series are shown, one for each sample regime. The sample regime series overlap during the period 2004 to 2014, allowing for an assessment of the effect of state entry/exit. During the overlap period all series have identical trends and any difference in levels appears to be directly related to the number and size of states omitted from each regime. Given the small share of missing employment it is not surprising Regime 3 overlays the complete data Regime 2.

Comparing Regime 1 with Regime 2, we observe a slight difference in volatility levels for the Regime 1 DLE volatility measure relative to the Regime 2 DLE volatility measure from 2004 to 2009, although in practice the difference is unlikely to significantly affect inferences. The differences across regimes for the APC measure are even smaller, suggesting much of the

difference in volatility across regimes is due to large changes in earnings, which typically occur for workers with very small earnings in one year of the year-pair. For all practical purposes earnings volatility estimated using a composite sample composed of the largest available regime each year is virtually identical to results estimated separately for each regime. Therefore, going forward all LEHD estimates use the composite sample.

Figure 1 - Variance of the Difference in Log Earnings (DLE) and Arc Percent Change (APC) by Sample Regime and Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. LEHD data has state entry/exit during the analysis period. Sample regimes are constructed such that sets of states have the same entry/exit dates. Regime 2 (R2) contains all 50 states plus DC. Regime 1 (R1) does not include AL, AR, DC, DE, IA, MA, MS, NE, NH, OK, UT, VT. Regime 3 (R3) does not include AK, MO, and SD.

In Figure 1 we have a preview of our baseline results (which are also shown in Figures 2 and 3 for the composite sample), setting the reference point for the rest of the paper. Excluding business cycle effects, earnings volatility has declined over the analysis period, a finding robust to various sensitivity tests. A similar result is present for both the DLE or APC measures of earnings volatility, although the business cycle effects are more pronounced, as expected, when using the unbounded DLE measure. We can also see two distinct periods, the period before the great recession where there is almost no trend in earnings volatility and the post great recession

period where we have a long decline in volatility to levels below those seen at the beginning of the series.

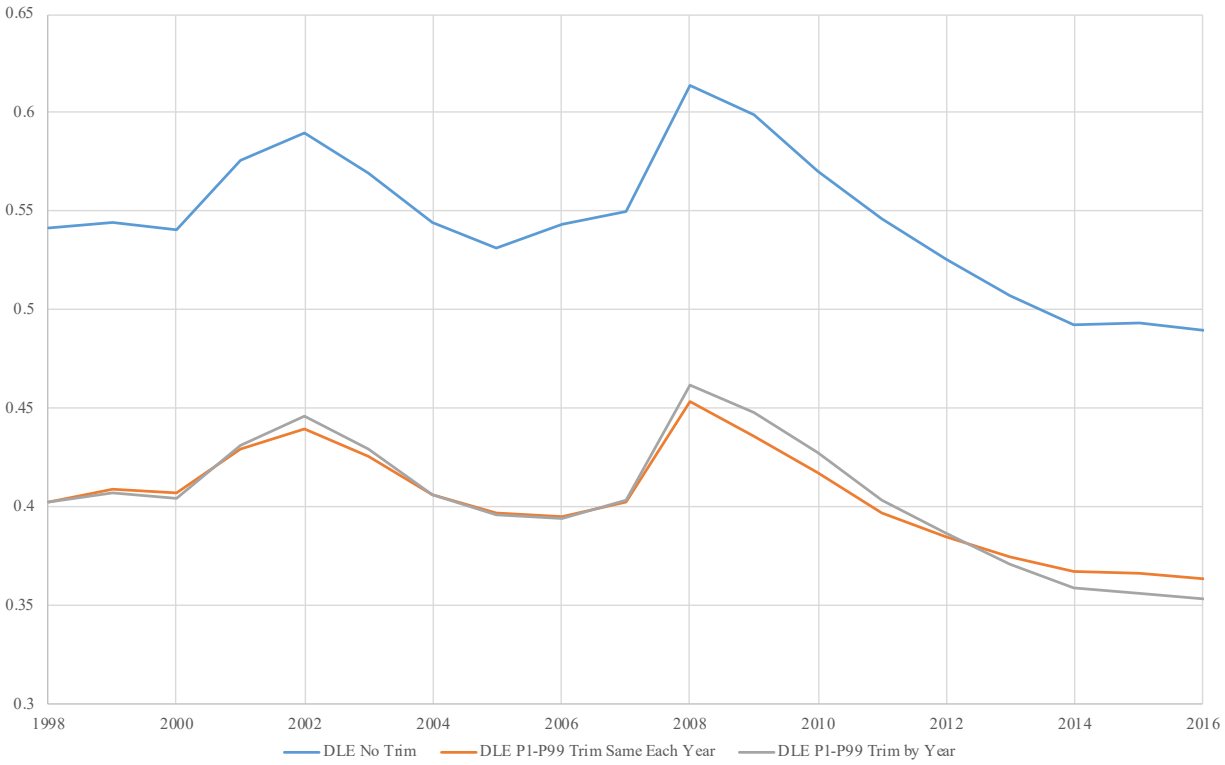
b. Earnings Distribution Trimming

Starting in this section, and continuing through the rest of the paper, we conduct a large number of sensitivity tests. Our first sensitivity test concerns trimming methods. Standard practice in the literature is to separately trim initial and subsequent year earnings using stated minimum and maximum values, thus placing an upper bound on the absolute value of the earnings change. We call this a "by-year" trim. A secondary but perhaps more important effect of trimming is that it reduces the number of very low earning workers, for whom a relatively small absolute change in earnings can result in a large percentage change. We implement trimming by year, setting the minimum at the 1st percentile and the maximum at the 99th percentile of the real earnings distribution for the actual data year. We also implement constant trimming using the same percentiles estimated from the combined earnings distribution in all years of the composite sample. There is merit in both constant and by-year trimming. By using the same values each year, constant trimming prioritizes a worker's absolute position in the real earnings distribution, which can be especially important for workers at the bottom of the earnings distribution where the absolute level of earnings may be material to their existence. However, using time-varying trim values allows us to include the set of workers each year who are in the same relative position in the real earnings distribution.

The choice of trim values is especially important for trend analysis if there are large changes in the tails of the cross-sectional earnings distribution over time. As we show in detail in section III.g., the relative contribution of workers in a specific part of the earnings distribution to earnings volatility increases as either the share of workers in that part of the earnings distribution increases, relative volatility for the typical worker in that part of the earnings distribution increases, or some mixture of the two. Trends estimated using either the constant trim or the by-year trim would be similarly affected by increases in relative volatility in the tails of the earnings distribution, however, trends estimated using a by-year percentile based trim will likely differ from trends estimated using a constant trim if the share of workers in the tails is changing over time. The percentile based trim values will shift down as the share of workers in

the tails increases allowing more low earning high volatility workers into the sample then the constant trim, increasing overall earnings volatility.

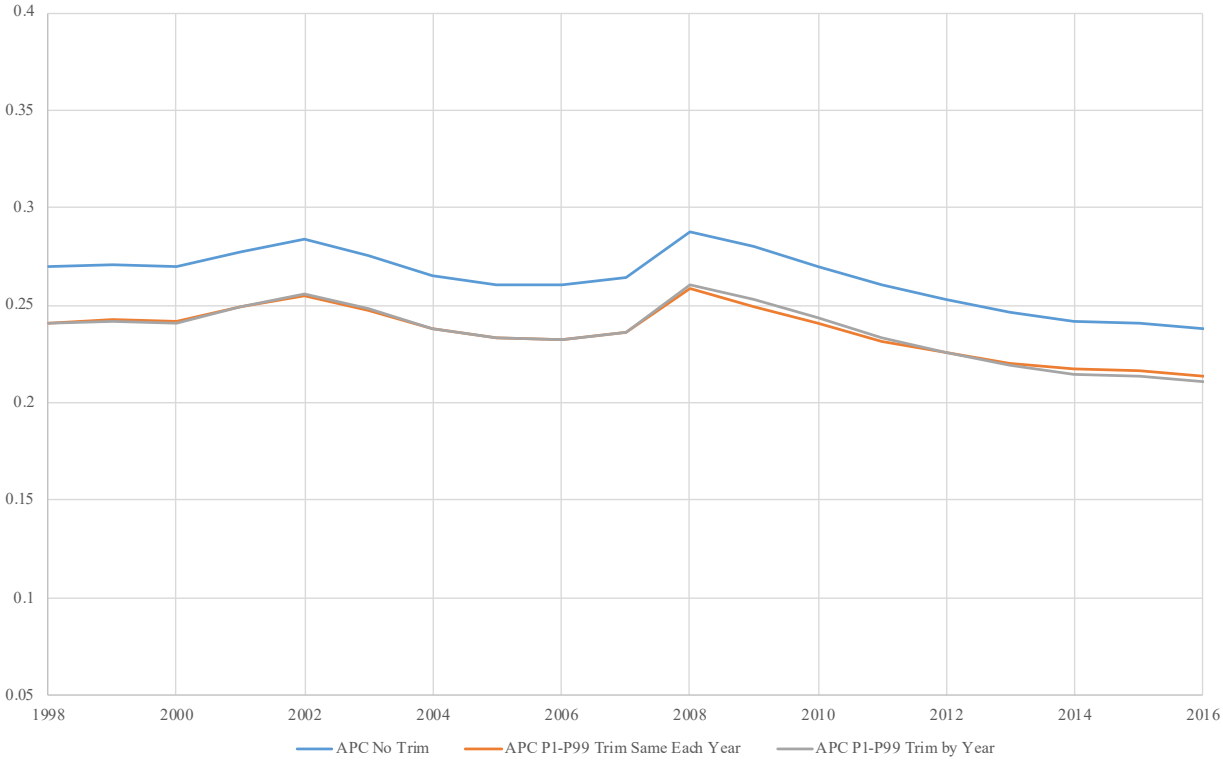
Figure 2 - Variance of the Difference in Log Earnings (DLE) by Trim Type and Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. Trim P1 ~ $300 and trim P99 ~ $300K.

The results for the DLE measure of volatility are shown in Figure 2 and the results for the APC measure of volatility are shown in Figure 3. The no-trim series in Figures 2 and 3 are the same series shown in Figure 1, but they are estimated from the composite sample. Three themes are evident. First, earnings volatility is counter-cyclical with increases during recessions and decreases during expansions. Second, ignoring the cyclicality, earnings volatility is relatively stable until 2008, when it begins a sustained decline. Third, volatility measured using the difference in log earnings is always larger than volatility measured using the arc percentage change.

Figure 3 - Variance of the Arc Percent Change (APC) by Trim Type and Year



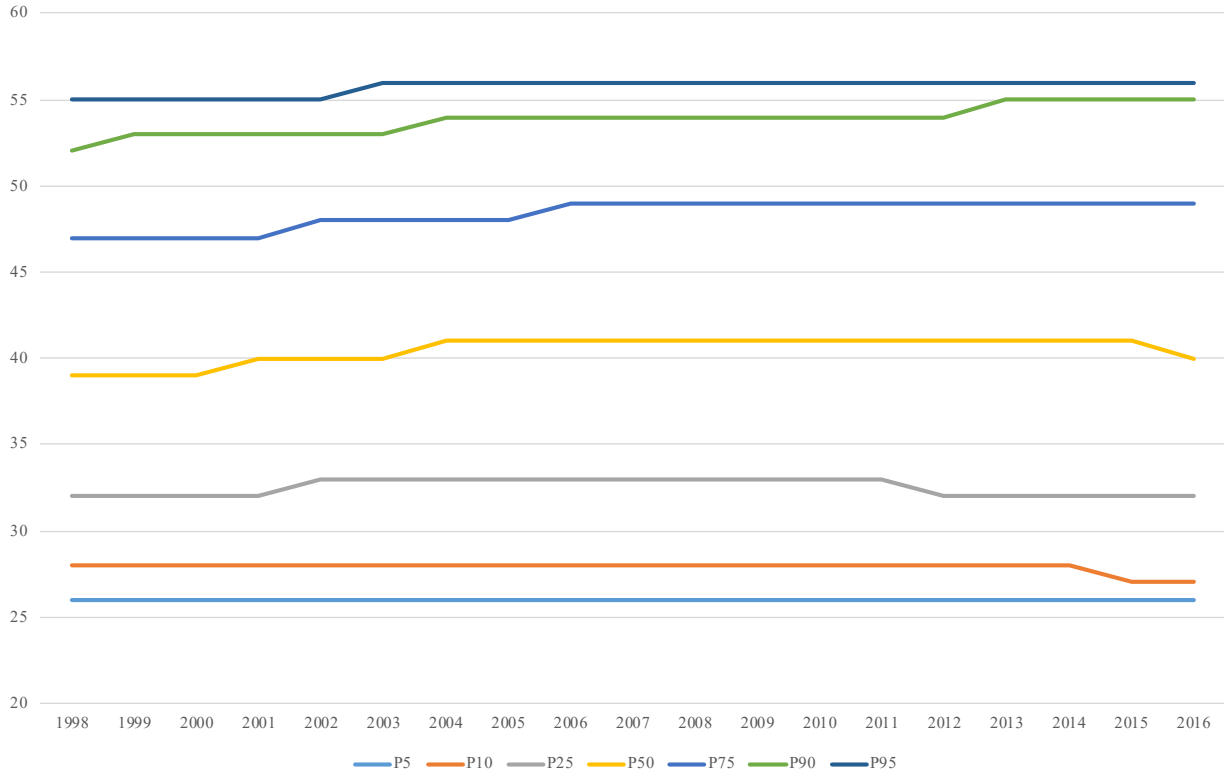
Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. Trim P1 ~ $300 and trim P99 ~ $300K.

Not surprisingly, the primary effect of trimming is to reduce the level of measured volatility. For example, using the DLE measure the untrimmed level of volatility at the 2008 peak of the great recession is about 0.62, while the trimmed values are 0.45 for the constant trim and 0.46 for the by-year trim. The substantial reduction in volatility highlights the sensitivity of the variance to the tails of the earnings distribution. The trends for the trimmed and untrimmed series are similar, although there are small, but noticeable, differences between the constant-trim and by-year trim estimates. Volatility is larger in the by-year trim series during recessions as workers face negative earnings shocks and the distribution shifts to the left, resulting in a larger share of lower earnings workers in the analysis sample and higher volatility. The results for the arc percentage changes are similar, but the effects of trimming are smaller. The arc percentage change already applies a smooth limit to the volatility calculation, reducing the impact of the trim. For the same reason, the increase in volatility during recessions is also muted, irrespective of the trim.

c. Age Distribution

Ignoring increases in volatility during recessions, earnings volatility declined from 1998 to 2016. One plausible explanation for the decrease in volatility is a shift in the age distribution to the right and/or increased labor force participation among older workers. If older workers have higher earnings and more stable employment this would generally imply a decrease in earnings volatility. Figure 4 shows the age distribution for active workers by year. The age distribution below the median is mostly stable over time, however there is a mild shift to the right in the age distribution above the median. For example, in 1998 the 75th percentile has an age of 47 compared with 49 in 2016.

Figure 4 - Age Distribution by Year



Notes: Modal age for males age 25-59 during the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year with annual earnings at all jobs between lower (P1) and upper (P99) trim values. Trim P1 ~ $300 and trim P99 ~ $300K.

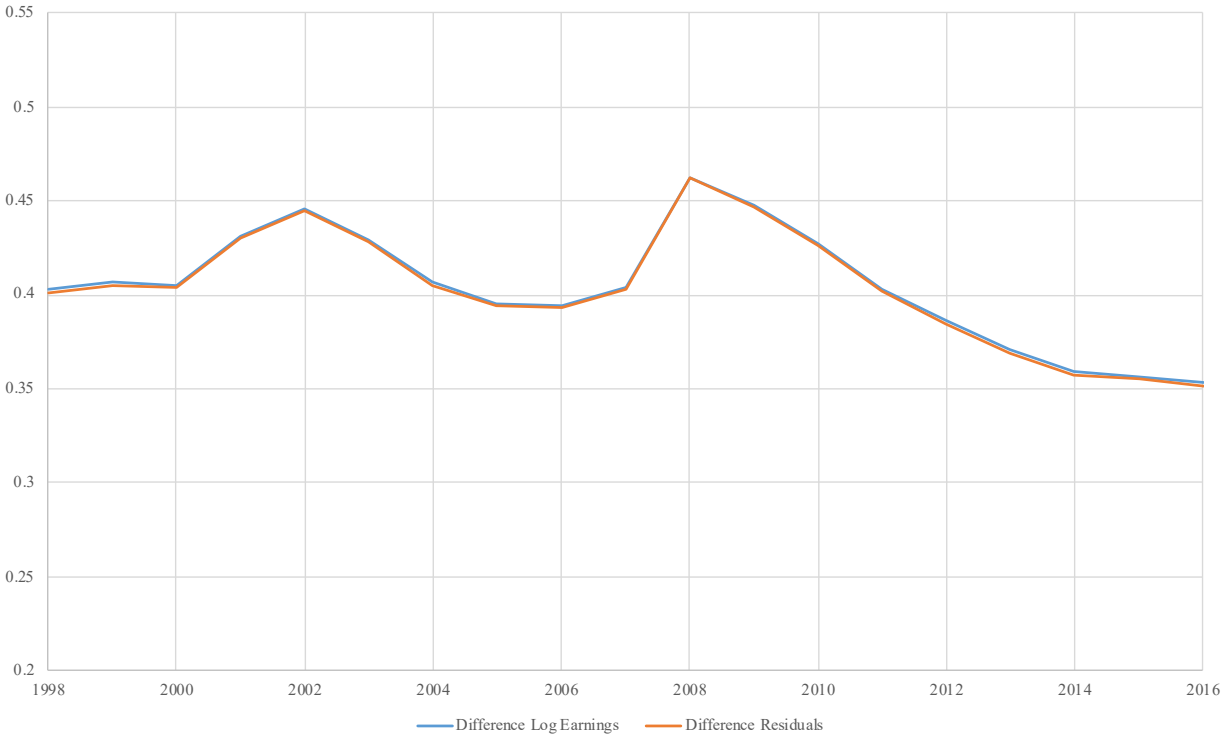
Although a shift in the age distribution is suggestive, for our purpose we need to determine if a shift in the age distribution and/or the age-earnings profile has a significant effect

on earnings variability. To test this hypothesis, we estimate the following regression model each year, where $y_{it} = \ln(e_{it})$ (see Shin and Solon 2011 for an early example of this approach).

$$y_{it} = \beta_{0t} + \beta_{1t} age_{it} + \beta_{2t} age_{it}^2 + \varepsilon_{it}. \quad (4)$$

In the next step we use the estimated age-adjusted residuals $\hat{\varepsilon}_{it}$ to calculate the difference in the age-adjusted log real annual earnings residuals $r_{it} = \hat{\varepsilon}_{it+1} - \hat{\varepsilon}_{it}$. The results of this exercise are shown in Figure 5, where we clearly see the variance of the difference in the age-adjusted residuals is virtually identical to the variance of the difference in log earnings.[5]

Figure 5 - Variance of the Difference in Log Earnings and the Difference in the Age Adjusted Log Earnings Residuals by Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Each worker must have reported earnings in both the initial and the subsequent year. Trim P1 ~ $300 and trim P99 ~ $300K. Residuals are from by year regressions of log real annual earnings on a constant, age, and age-squared. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016.

[5] We also estimated a more general form of the age adjustment directly on the difference in log earnings with similar results.

The result is not surprising. When we difference log earnings, we implicitly remove the age effect except under very specific conditions. The next two equations show the age adjusted log earnings regression equations in year t and $t+1$:

$$y_{it} = \beta_{0t} + age_{it}\beta_{1t} + age_{it}^2\beta_{2t} + \varepsilon_{it} \tag{5}$$
$$y_{it+1} = \beta_{0t+1} + (age_{it} + 1)\beta_{1t+1} + (age_{it} + 1)^2\beta_{2t+1} + \varepsilon_{it+1}.$$

Taking the difference $(y_{it+1} - y_{it})$ and grouping terms gives the following relationship between the difference in log earnings and the difference in the residuals.

$$\begin{aligned} y_{it+1} - y_{it} = {} & age_{it}(\beta_{1t+1} - \beta_{1t}) + age_{it}^2(\beta_{2t+1} - \beta_{2t}) \\ & +2age_{it}\beta_{2t+1} + (\beta_{0t+1} - \beta_{0t}) \\ & +\beta_{1t+1} + \beta_{2t+1} + (\varepsilon_{it+1} - \varepsilon_{it}) \end{aligned} \tag{6}$$

First, any terms constant for all i are absorbed by the mean and will not affect the variance. Second if the age-earnings profile is the same in adjacent years, then the first two terms are zero and the gap between $(y_{it+1} - y_{it})$ and $(\varepsilon_{it+1} - \varepsilon_{it})$ is linear and increasing with age. The amount the gap increases with age is determined by $2age_{it}\beta_{2t+1}$, which reflects the concavity of the age-earnings profile. If the age-earnings profile is both linear and identical in adjacent years, then the variance of the difference in log earnings will exactly equal the variance of the difference in the residuals. Only if the age earnings profile is changing rapidly over time and/or if the age-earnings profile is strongly concave will we see a noticeable difference in volatility between the two measures. Although the estimated age profiles are concave, the estimated coefficients for β_{2t} and β_{2t+1} are small and the age-earnings profiles are relatively stable from year-to-year resulting in virtually identical estimates of volatility using either the difference in log earnings or the age adjusted log earnings residuals.[6]

d. Labor Force Attachment

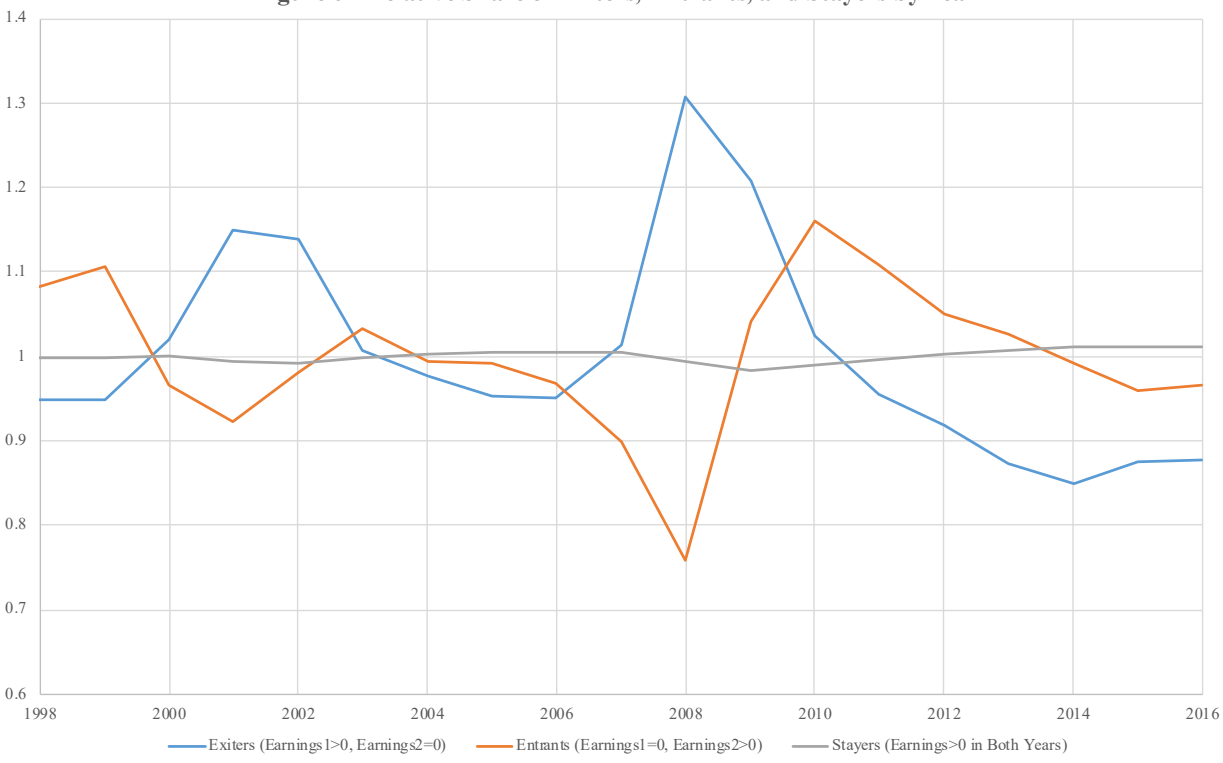
Up to this point, we have focused on workers with reported earnings in two consecutive years, however many workers are inactive for relatively long periods of time, especially during

[6] The year-to-year magnitude of the changes in the age-earnings profiles is what is material to our analysis. Somewhat counterintuitively, a gradual shift in the age-earnings profiles will likely not affect our results, however, a high level of time varying short duration changes in the age-earnings profile would be a concern.

recessions. Our annual earnings measure is based on calendar year earnings; therefore, there is not a direct relationship between observed annual earnings and long inactivity spells. Depending on the start and end dates of the worker's job, a worker with a similar gap in employment may have earnings in two adjacent years while another worker may not. Although we cannot resolve this timing issue, we can increase our capture of long duration periods of inactivity by allowing for year-pair observations where a worker has positive reported earnings in only one of the two years.

We label workers with reported earnings only in the first year as "exiters," and workers with reported earnings only in the second year as "entrants." Workers with reported earnings in both years are labeled as "stayers." In a typical year (composite sample average) approximately 89% of workers are stayers, 6% of workers are exiters, and 5% of workers are entrants. Graphs of the shares of exiters, entrants, and stayers by year relative to the sample averages are shown in Figure 6.

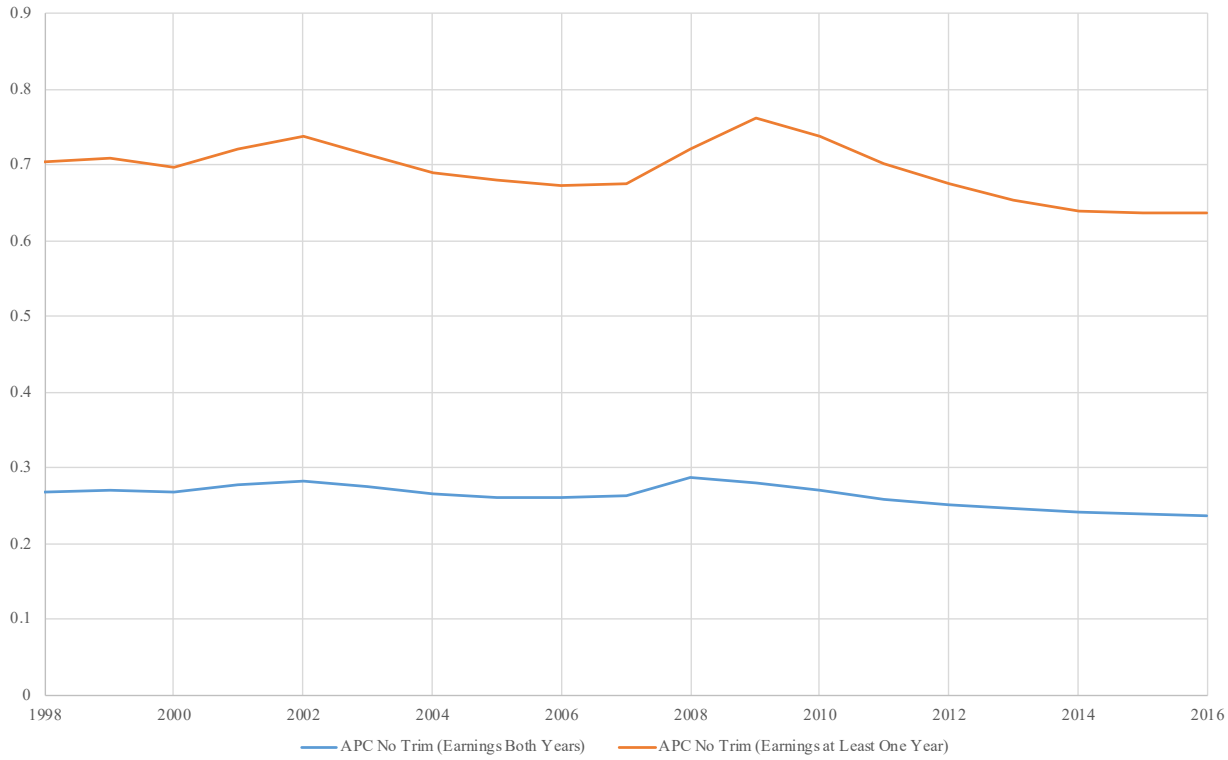
Figure 6 - Relative Share of Exiters, Entrants, and Stayers by Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in either the initial and/or the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. Shares are relative to the average share across all years. Average share of exiters is 0.062. Average share of entrants is 0.053. Average share of stayers is 0.885.

The recessions in 2001 and 2008 were both associated with large changes in the share of exiters and entrants. During recessions, exiters increase and entrants decline, and there are small changes in the share of stayers. During the 2001 recession the elevated share of exiters and reduced share of entrants begins in 2000 and ends in 2003. During the 2008 recession the divergence begins in 2006 and ends in 2009, although both exiters and entrants are noticeably elevated in 2009 relative to 2003. The average share of exiters being greater than the average share of entrants implies the flow of exiters is greater than the flow of entrants and that net exiters (exiters – entrants) is positive. The dip in the share of stayers is also noticeable during the great recession, especially given their large share. Although entrants are elevated from 2009 to 2013 and exiters are depressed from 2011 forward, employment does not fully recover from the great recession until 2014. Finally, we also observe less mobility into and out of active status from 2014 forward with more stayers, fewer exiters, and fewer entrants.

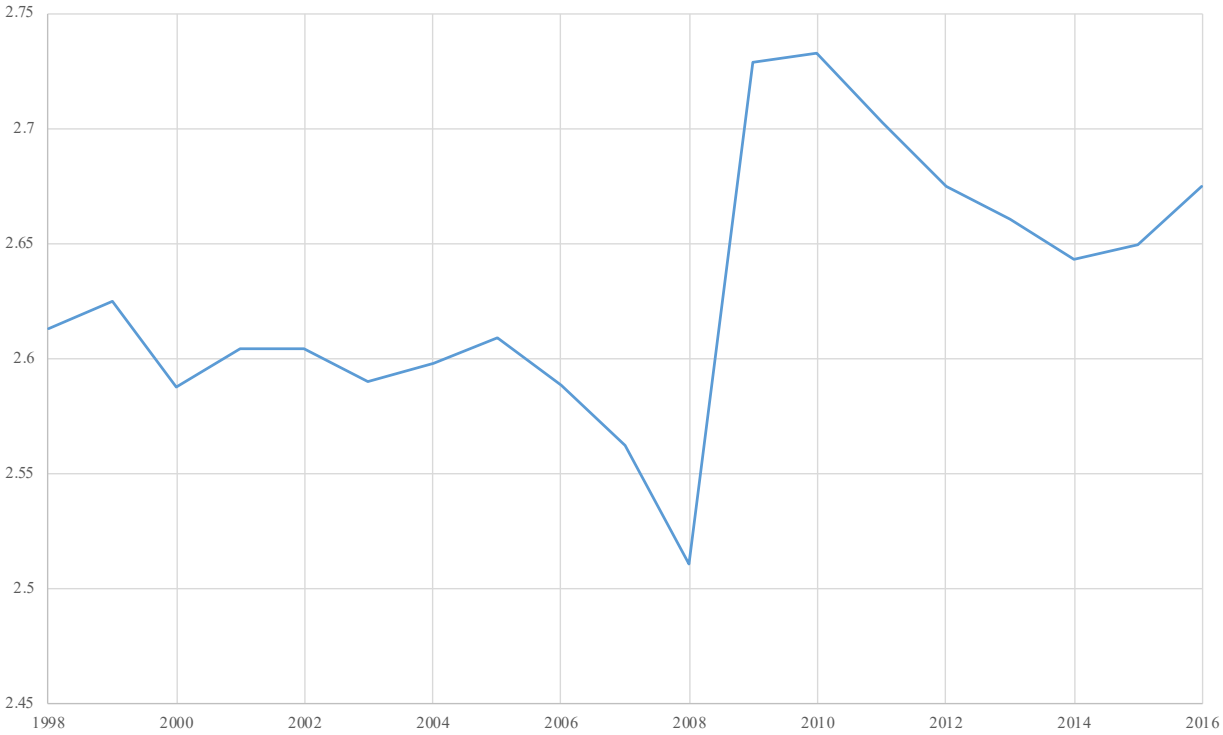
Figure 7 - Variance of the Arc Percent Change (APC) by Year Including Zero Earnings



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in either the initial and/or the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016.

In Figure 7 we show series for the arc percentage change—one including entrants and exiters, and the other excluding them. The results highlight the significant share of earnings volatility missed by not including exiters and entrants. Including workers with earnings in only one year of the year-pair substantially increases estimated earnings volatility—from about 0.27 to 0.70 in year 2000, for example. As predicted from our discussion of Figure 6, the peaks are larger during recessions as entrants and exiters increase, while stayers decrease. Although it may not be clear from Figure 7, the relationship between the two APC volatility measures appears to undergo a structural change in the latter half of the time series. Figure 8 plots the ratio of the APC including zeros to the APC excluding zeros. From 1998 to 2006 the relationship is relatively stable. However, prior to the great recession the APC including zeros decreases from its average value of 2.60 to 2.52 before skyrocketing to 2.73 at the peak of the great recession. Curiously, even after employment has fully recovered from the recession in 2014, the ratio does not decline to its previous level of 2.60 but fluctuates at values of 2.64 or higher. This suggests that, in spite of their historically low levels from 2014 forward, entrants and exiters make up a larger share of volatility from 2009 forward than they did prior to the great recession, implying that volatility for stayers is lower, an issue we revisit later in the paper.

Figure 8 - Ratio of the Variance of the Arc Percent Change Including Zeros to the Arc Percent Change Not Including Zeros



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in either the initial and/or the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016.

e. Symmetry

Although the economic impacts of *real* earnings increases differ dramatically from those of earnings decreases, neither of our measures of volatility distinguish between the two.[7] In this section we endeavor to understand potential changes in the share of volatility explained by real earnings increases relative to earnings decreases. [8] To disentangle the effects of workers with real earnings increases from those with real earnings decreases we use the following decomposition of the total variance

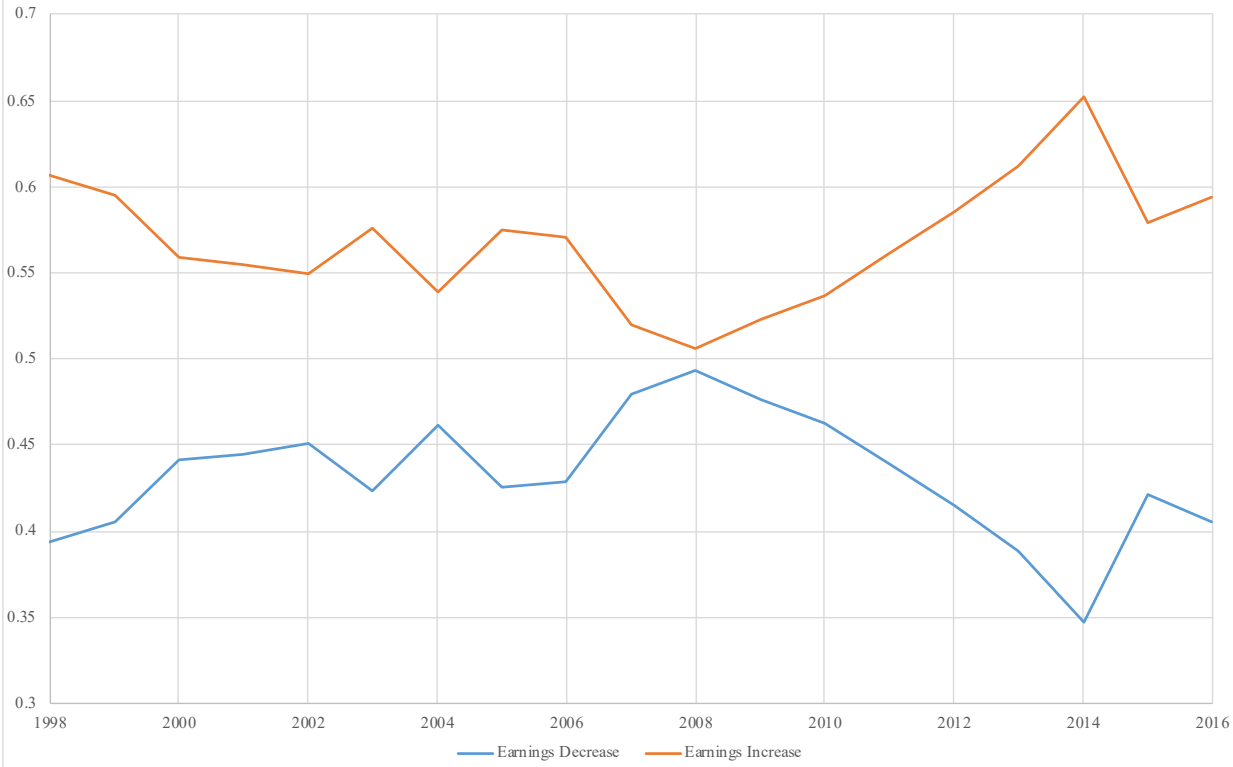
$$Var(x) = p \times var(x_1) + (1-p) \times var(x_2) + p \times (1-p) \times \big(\mu(x_1) - \mu(x_2)\big)^2. \qquad (7)$$

[7] We measure real earnings changes for workers with earnings in both years of the year pair. Some workers with small real earnings decreases may have nominal earnings increases. Workers classified as having a real earnings increase includes workers whose real earnings have not changed.

[8] Please see Appendix A for estimates of the distribution of the change in log real annual earnings by year. The results in Appendix A do not alter the theme of the results presented in this section but provide additional context regarding the changes discussed here.

The formula shows how a given earnings volatility measure x, where x is either the DLE or the APC, can be decomposed into a weighted average of the within variances for each group (in this case group 1 is persons with increasing earnings and group 2 is those with decreasing earnings) plus a mean difference component. When the means of the change in earnings in two sub-populations are equal, $\mu(x_1) = \mu(x_2)$, the total variance is the weighted sum of the sub-population variances. The total variance fluctuates when the variance of each group changes, the share of workers in each group changes, and/or the mean difference ($\mu(x_1) - \mu(x_2)$) changes.

Figure 9 - Share of Workers With an Earnings Decrease (Increase) by Year

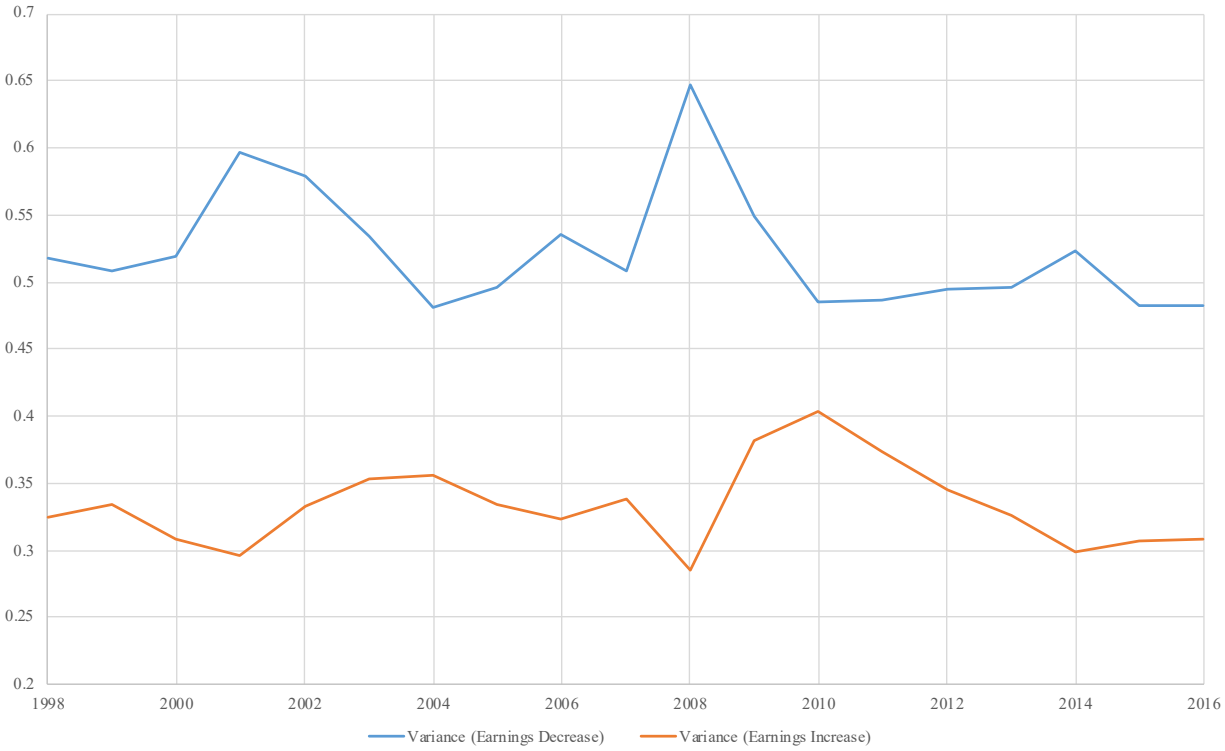


Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016.

Applying the decomposition to our two groups, Figure 9 shows the share of workers with earnings increases and earnings decreases over time. The share of workers with year-to-year real earnings increases declines gradually from 1998 to 2008 and then increases strongly during the

recovery from the Great Recession, peaking in 2014 before declining to 1998 levels in 2015/2016. The graph for earnings decreases is necessarily the mirror image. Depending on the within-variance of workers in each group $var(x_1)$ and $var(x_2)$, the changes in the shares may provide a potential pathway for explaining changes in the total variance $var(x)$.

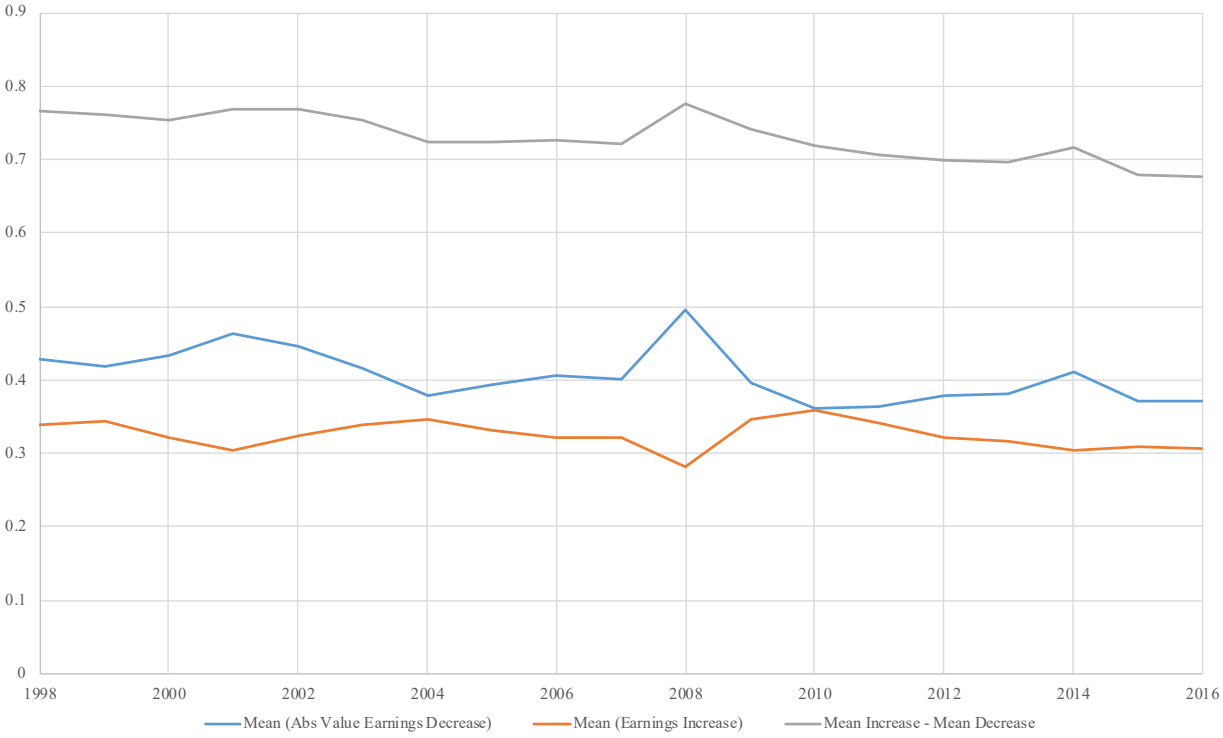
Figure 10 - Variance of the Difference in Log Earnings for Workers With an Earnings Decrease (Increase) by Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016.

Figure 10 shows the time series of the within variance for each group. The variance of the difference in log earnings for workers whose earnings decrease is larger than the variance for workers whose earnings increase. This implies that as the share of workers whose earnings increase goes up, the total variance declines. Figure 11 shows the mean difference in log earnings for workers in each group. The difference in mean earnings for each group is large (around 0.7) and therefore changes in the mean difference will also have an important impact on total volatility.

Figure 11 - Mean Difference in Log Earnings for Workers With an Earnings Decrease (Increase) by Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016.

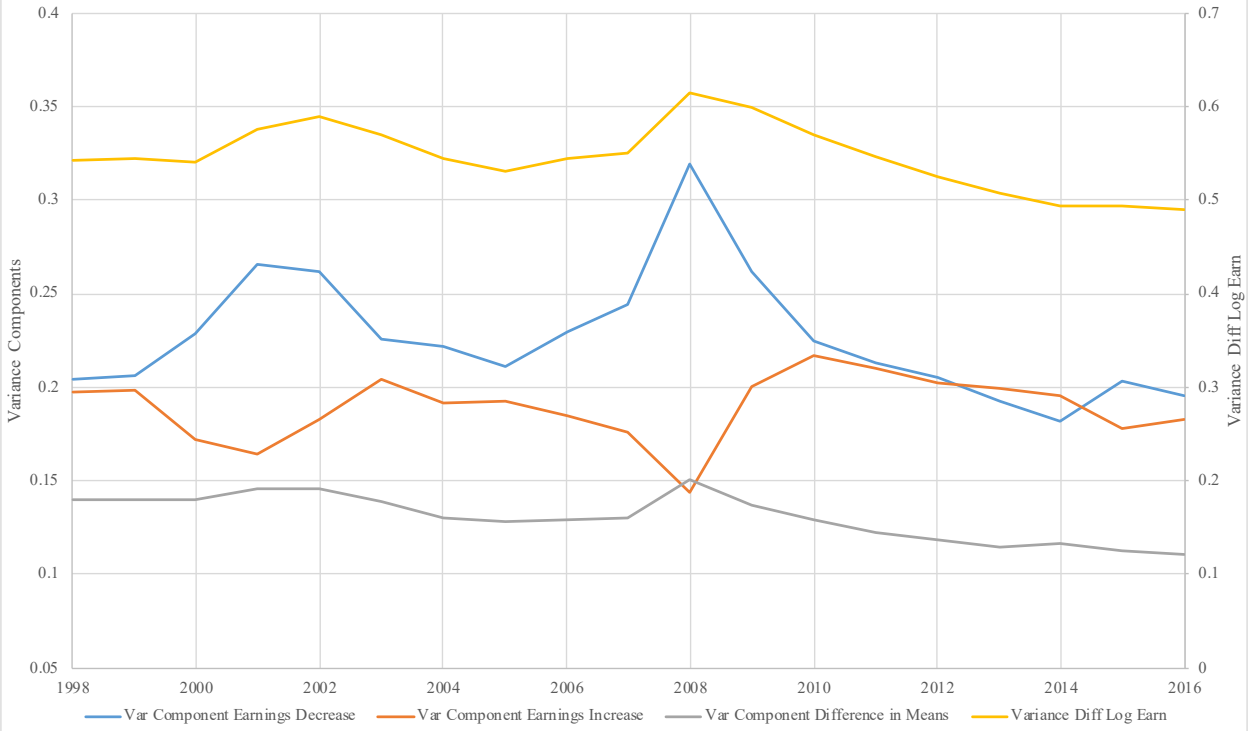
Since differences in the within-group mean and the variance of the log earnings both impact the overall distribution, we discuss them together. For workers whose earnings decrease, there is a general downward trend in both the variance and the absolute value of the mean difference in log earnings.[9] Both components reduce this volatility measure—a worker whose earnings decrease in 2015 compared with 2000 is likely to have both a smaller average log earnings decrease (in absolute value) and less risk of a large deviation from that average. For workers whose earnings increase, the downward trend in within-group mean and variance also implies a decrease in log earnings volatility (especially for 2012 forward) because the smaller increases are more closely clustered around the mean.

Figure 12 combines the three components together, showing how overall earnings volatility changed over the analysis period. The first two variance components in Figure 12 are

[9] The mean earnings for workers with a decrease is negative. We plot the absolute value to facilitate comparison with other components.

the within-variances each multiplied by their respective share. The mean difference component is the square of the difference in means multiplied by the share times one minus the share $\left(p \times (1-p) \times \left(\mu(x_1) - \mu(x_2)\right)^2\right)$. The figure shows that the total variance of the difference in log earnings is relatively flat, except during recessions, until 2008, when it begins a long decline. During a recession, the variance of negative earnings shocks increases while the variance for workers with positive shocks declines. At the same time, the share of workers with negative shocks increases, further increasing total variance. The difference in means also contributes to the increase in total variance during a recession, because as the difference in means increases total variance increases. During the recovery, the reverse happens.

Figure 12 - Earnings Decrease (Increase) Components of the Variance of the Difference in Log Earnings by Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. The variance components sum to the variance of the difference in log earnings.

The decline in total earnings volatility from 2010 forward involves changes in all of the components, each contributing to a decline in earnings volatility. The variance component for workers whose earnings increase is declining, although there is a large increase in the share of

workers whose earnings increase, which has the effect of dampening the decrease in this component of total variance. However, the converse is true for the variance component for workers whose earnings decrease. This component is declining primarily due to the reduced share of workers whose earnings decrease. The mean component is also declining due to a share effect (more unequal distribution of workers between groups) and a difference in mean effect (the rate of decline in mean earnings increases outpaces the rate of increase in the absolute value of earnings decreases). Overall, the consequences for the typical worker of the overall decrease (post great recession) in earnings volatility are mixed. On the one hand, the share of workers whose earnings increases is relatively high, however the increases are smaller on average and the variance is low (potentially limiting upward mobility).

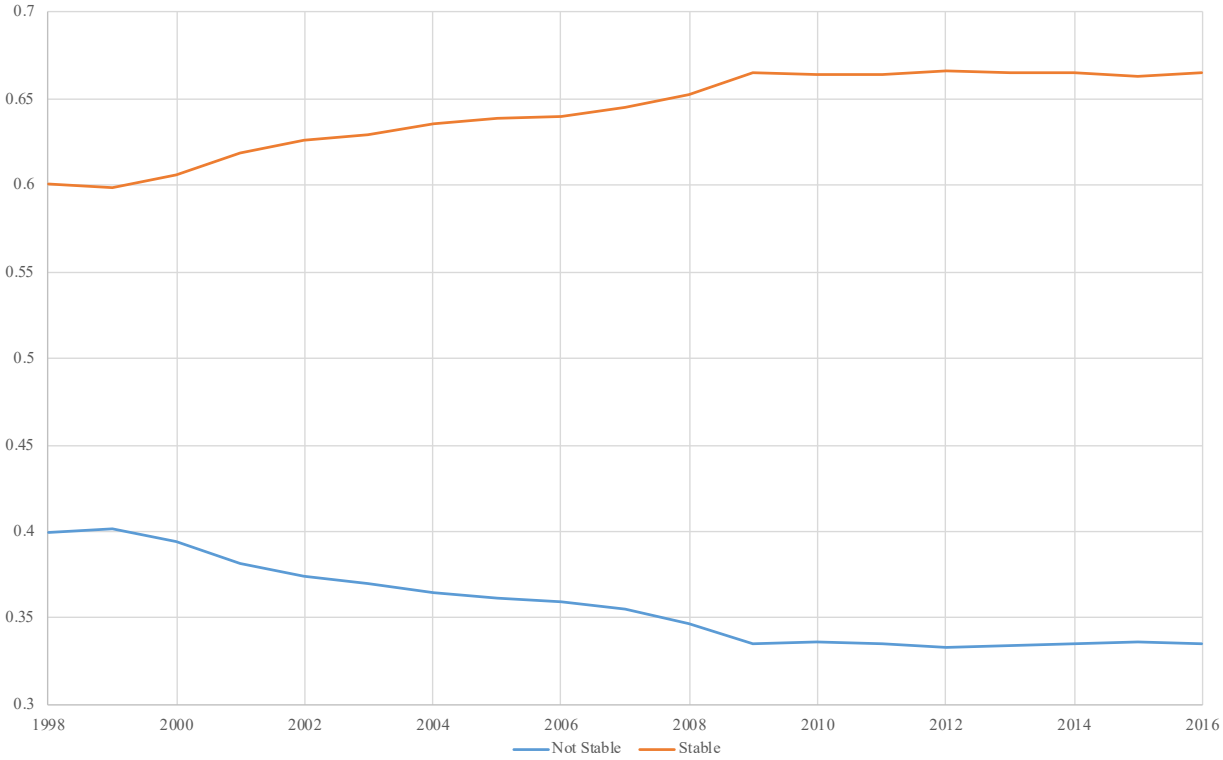
f. Stability

Although this paper's primary focus is earnings volatility, paradoxically the majority of workers do not experience large year-to-year changes in earnings.[10] Earnings volatility is driven primarily by the smaller set of workers that experience a large shock in labor supply and/or change employers. Using the techniques from the previous section, we separate workers into two groups: stable and not stable. Stable workers are defined as receiving the majority of their earnings in both years of the year-pair from the same (dominant) employer.[11] In addition, we require stable workers to be active all eight quarters of the year-pair. Every worker that does not meet these criteria and is active at least one quarter each year is labeled as not stable. The stable worker definition is deliberately designed to be selective, workers must not change the employer that provides the majority of their earnings each year over a two-year period; however a stable worker may have multiple jobs and small breaks in active status.

Figure 13 shows the share of stable and not stable workers by year. Stable workers make up about 60% of all workers in 1998, increasing to about 67% in 2016. The share of stable workers increases almost continuously from 1998 to 2009, when the growth in the share of stable workers stops. Not coincidentally perhaps, the share of stable workers stops increasing during the Great Recession and is constant thereafter.

[10] Approximately 50% of workers have a change in earnings of less than +/- 4% from the initial to the subsequent year. Please see Appendix A for estimates of the distribution of the change in log real annual earnings.
[11] The dominant employer is the one from whom the employee receives the largest proportion of annual earnings during the year.

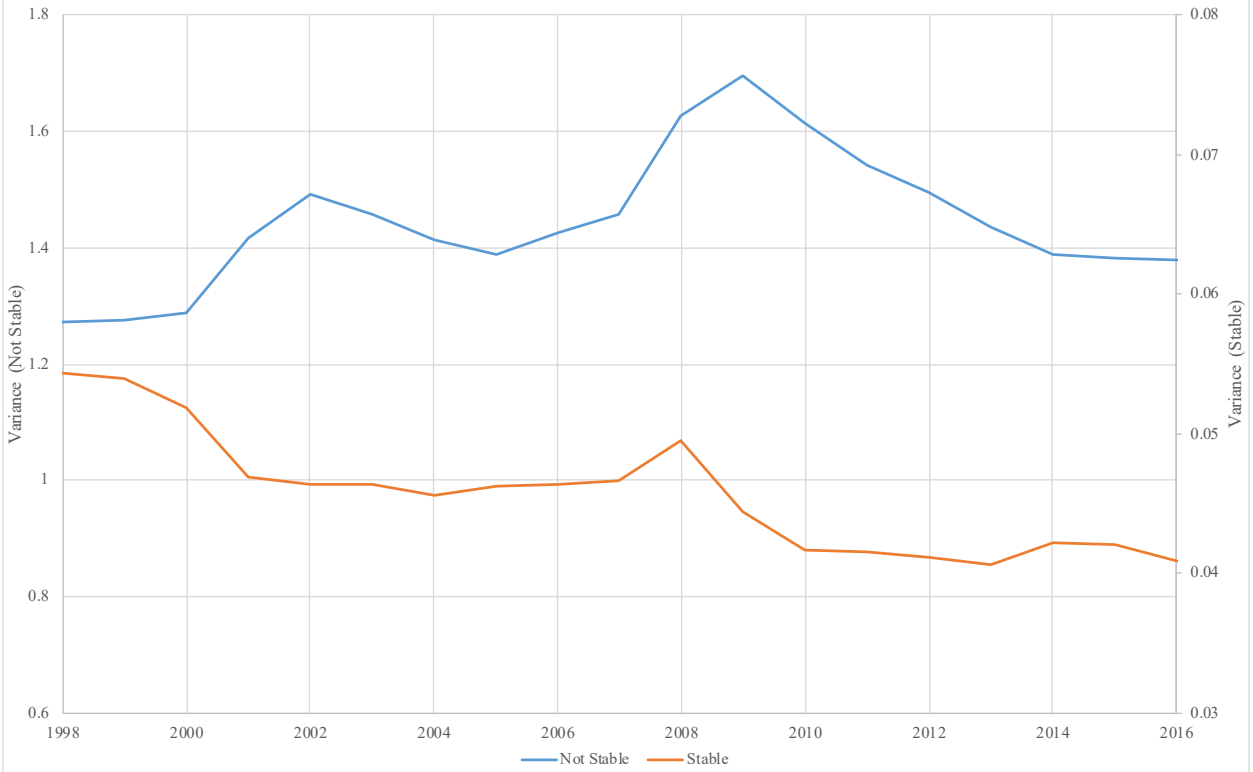
Figure 13 - Share of Stable and Not Stable Workers by Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. A stable worker has the same dominant employer in both years and is active all eight quarters of the year pair.

In Figure 14 we show the within-group variance for stable and not-stable workers each year. The two series tell very different stories. The variance of the difference in log earnings for stable workers is only a small fraction of the variance of not-stable workers (0.054 vs 1.27 in 1998). There are the different trends—earnings volatility for not-stable workers is increasing while earnings volatility for stable workers is decreasing. The impact of recessions on volatility is almost entirely born by not-stable workers. For example, earnings volatility for not-stable workers increases from 1.39 in 2005 to 1.70 in 2009 while volatility for stable workers declines almost imperceptibly from 0.046 to 0.044.

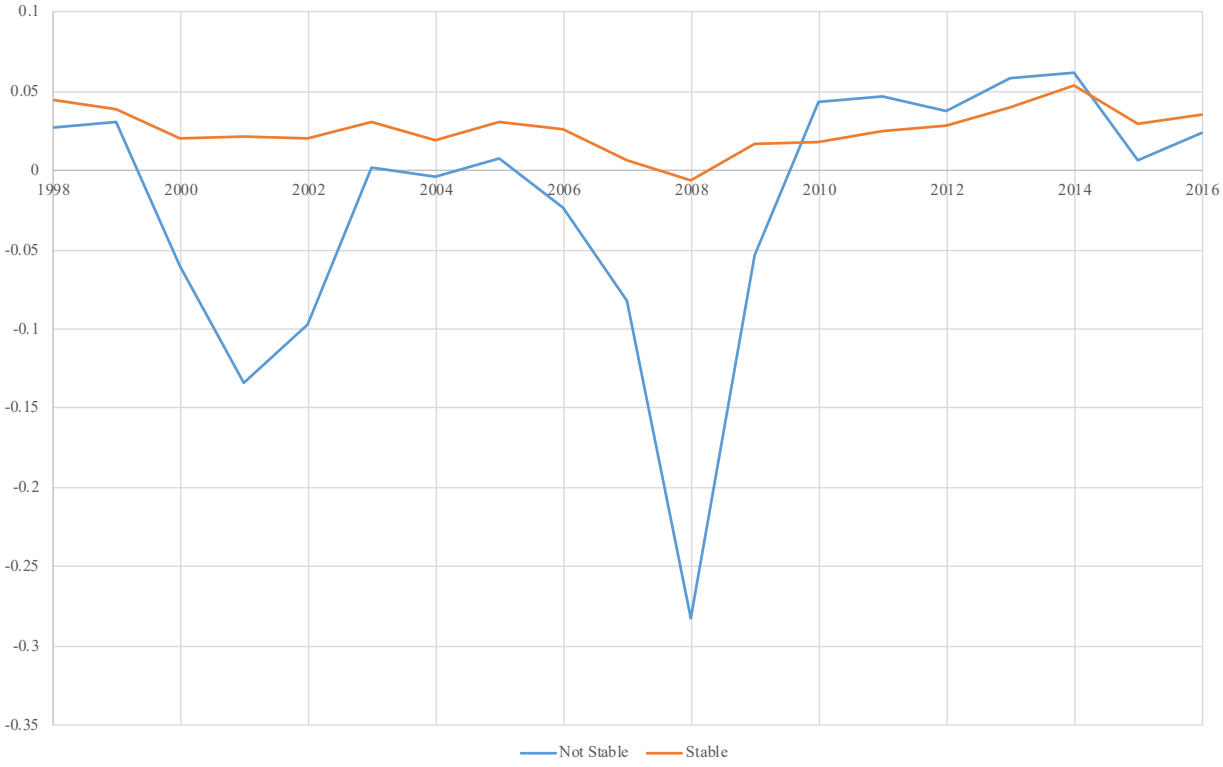
Figure 14 - Variance of the Difference in Log Earnings for Stable/Not Stable by Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. A stable worker has the same dominant employer in both years and is active all eight quarters of the year pair.

The differing experiences of stable and not-stable workers continue when comparing the mean difference in log earnings. Mean real annual earnings changes for stable workers are almost always positive (except for a small negative value in 2008), while earnings changes for not-stable workers are almost always negative (or barely positive) from 2000 to 2009. Workers consistently in the not-stable group fare especially poorly when we consider both the variance and the mean, with increases in earnings volatility added to the frequent large negative earnings shocks, results consistent with the increasing inequality that AMZ found between 2000 and 2011 in previous research using LEHD data. The only silver lining is that during the recovery from the Great Recession, the within-group variance for not-stable workers started a sustained decrease and the mean of the difference in log earnings turned positive.

Figure 15 - Mean of the Difference in Log Earnings for Stable/Not Stable by Year



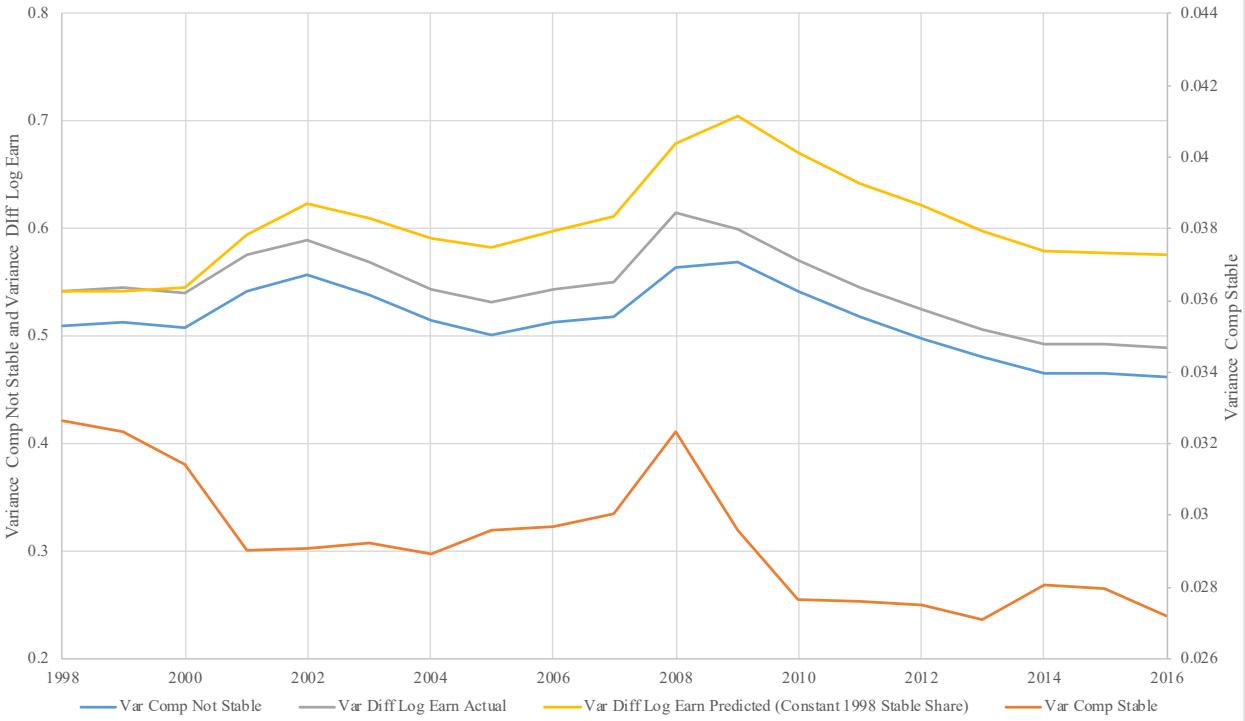
Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. A stable worker has the same dominant employer in both years and is active all eight quarters of the year pair.

These results paint a bleak picture for not-stable workers. Fortunately, and perhaps not surprisingly, from 1999 to 2009 workers have been moving into more stable employment. The shift of workers into more stable employment is the primary reason total earnings volatility has not increased (except during recessions) over the period. Figure 16 shows the results in more detail, highlighting the dominant contribution of not-stable worker volatility to the total as well as the important role of the increasing share of stable workers.[12] The variance component attributable to not-stable workers is typically responsible for about 93-94% of the total variance. Except for 2008, the not-stable group's variance component moves in virtual lockstep with the total variance of the difference in log earnings. The 60% of workers who are stable play only a minor role in determining the level of *total* earnings volatility, although given the large difference in the within-group variances, changes in the share can and do have a large effect. For example, if we held stable workers at their 1998 share, the variance of the difference in log

[12] The mean difference component is negligible and is not shown.

earnings would increase from 0.54 to almost 0.58 over the analysis period rather than decrease from 0.54 to 0.49 as observed, ceteris paribus.

Figure 16 - Stable and Not Stable Variance Components by Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. A stable worker has the same dominant employer in both years and is active all eight quarters of the year pair. The not stable and stable variance components are approximately equal to the variance of the difference of log earnings (actual). The mean difference component is negligible and is not shown.

g. Earnings Distributions

Estimated annual real earnings distributions differ depending on the specific data sources used, which can have a noticeable impact on estimated earnings volatility, especially if the differences are concentrated in the lower (left) tail of the earnings distribution. Our primary analysis dataset is based on state Unemployment Insurance (UI) records. Compared with survey data sources, UI data contain a relatively large number of low-earning workers. In this section, we document the differences in the earnings distribution between the LEHD, PSID, and a sample of LEHD earnings for workers who appear in the American Community Survey (the LEHD-ACS sample). We also show how differences in the estimated earnings distributions impact the

variance of the arc percentage change in earnings and whether differences affect trends in volatility.

LEHD earnings data are reported directly by the employer (often electronically) and have broad coverage of both the private sector and state/local government.[13] Earnings reports from household surveys take a less direct route to the analyst. To receive an accurate response the frame of households must cover the target population. If the household is selected, the respondent must both answer the survey and correctly answer detailed demographic and historical earnings questions for all in-sample persons in the household.[14] If there is a breakdown anywhere in the chain, the survey earnings reports may either be missing or inaccurate. For stable workers in a single-earner household, the respondent is both more likely to be the actual worker and have similar earnings each year, making it easier to recall previous-year annual earnings. However, in households with multiple workers and/or less stable employment histories there are many more challenges. First, the respondent is less likely to account for the majority of earnings in the household, potentially forcing that person to estimate earnings for other household members. Second, if the respondents have multiple employers with varying earnings from year-to-year it may be difficult to accurately recall previous year earnings. Third, even if the respondent has accurate earnings information for all jobs and for all in-scope persons, the survey may be designed to record earnings only from the primary or dominant job, resulting in an incomplete picture of total annual earnings.[15]

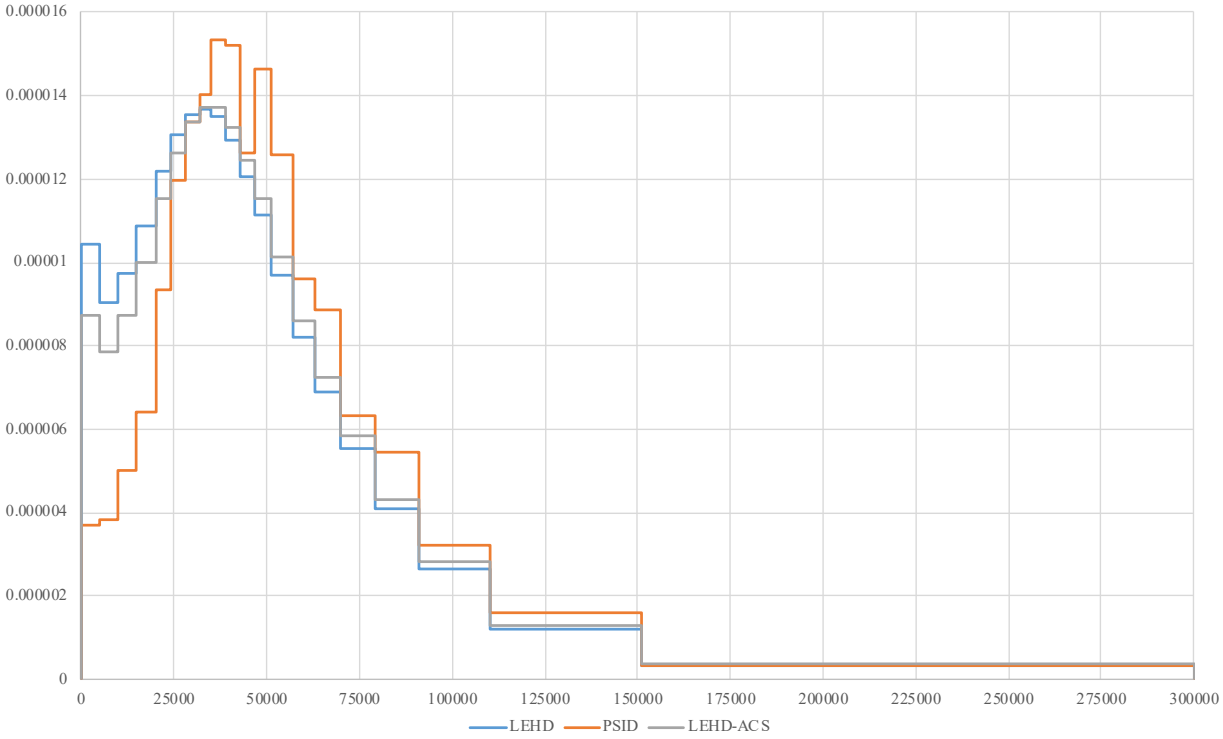
To better understand the magnitude of the differences in the earnings distributions across surveys we construct two additional samples, the first from the PSID and the second combining the workers sampled into the ACS with their LEHD earnings data. We use initial-year real annual earnings for workers with earnings in both the initial and the current year, pooling the years from 2002-2014. The PSID sample was constructed using the online PSID tool and is designed to match as closely as possible both the LEHD UI data sample (males, age 25-59) and the sample selection rules used in Moffit and Zhang (2020). The PSID is a biennial survey,

[13] See AMZ for an explanation of why we have excluded federal workers from this study.
[14] Both the ACS and the PSID ask one respondent per household to complete the data for all in-scope persons, although most researchers using the PSID typically only analyze earnings responses for the actual respondent or head of household.
[15] Obtaining accurate survey-based earnings responses can be challenging, although if the issues mentioned above affect data collection consistently over time then earnings volatility trends would be relatively unaffected. Consistency is the key when estimating trends, for example even procedures that improve data collection may affect the trend if they change the composition of the sample.

giving us potentially seven years of earnings for each respondent. The resulting analysis sample size is approximately 15,200 earnings observations. We constructed the LEHD-ACS sample by matching every person from the 2002 to 2014 ACS with that person's year-pair work history from the LEHD analysis sample used in this paper. The resulting LEHD-ACS analysis sample contains about 622 million paired earnings observations compared with 8.7 billion such observations in the composite LEHD sample.

We estimate the earnings distribution by calculating histograms for each data source. Each observation is placed into one of twenty bins with boundaries approximately equal to the ventiles of the 2002-2014 LEHD composite sample. Figure 17 shows the resulting earnings distributions for the three data sources. Relative to the LEHD and LEHD-ACS samples, the PSID earnings distribution is shifted to the right (except at the very top). The differences are especially pronounced for workers with earnings below $25,000. The LEHD data has about 25% of workers with earnings less than $25,000 and the LEHD-ACS sample has about 22%, while the PSID has only 13%. The results for the LEHD-ACS sample suggest that the ACS household frame and the persons reported in each household produce a worker population very similar to the one in our eligible-worker LEHD data. Most of the discrepancies found in AMZ between the ACS and LEHD data appear to be due to responses to the earnings questions, not differences between the ACS household frame and our eligible-worker frame.

Figure 17 - LEHD, PSID, and LEHD-ACS Earnings Distributions Pooled 2002-2014

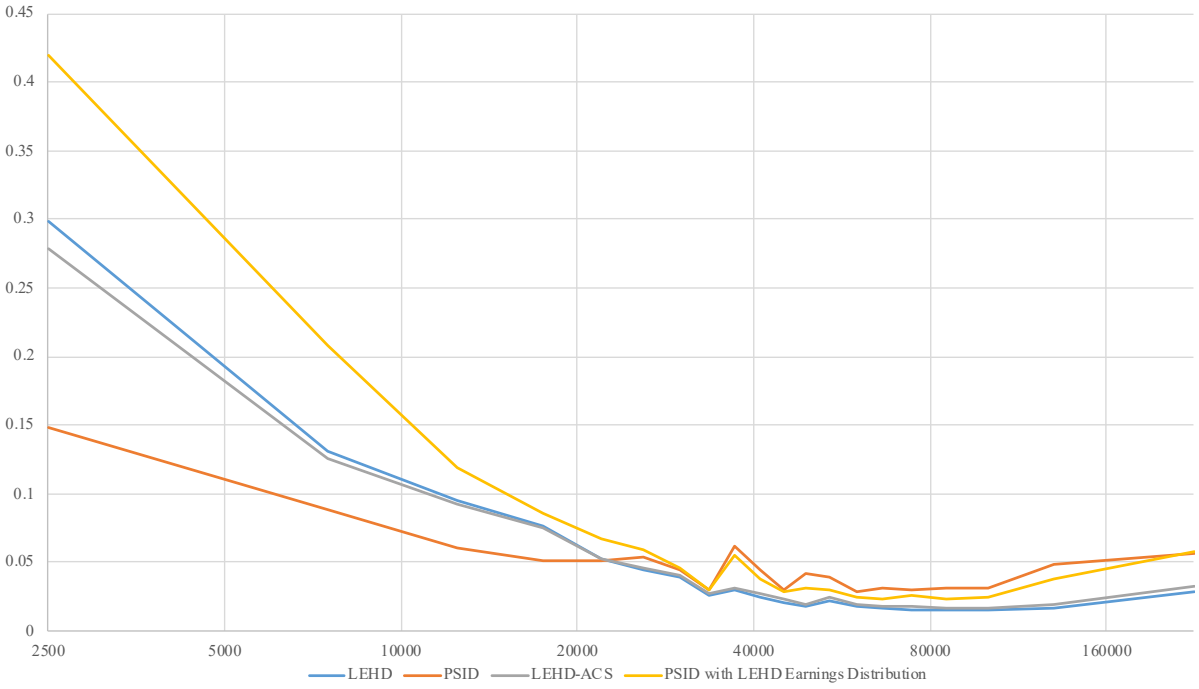


Notes: Real annual earnings (2010 PCE) at all jobs for males age 25-59. Each worker must have reported earnings in both the initial and the next available year (subsequent year for LEHD and LEHD-ACS, initial year+2 for PSID). Composite LEHD and LEHD-ACS sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014. The LEHD-ACS sample consists of LEHD annual earnings for all in sample LEHD workers that appear on the ACS during 2002-2014. The PSID sample includes biennial reports of annual wage and salary earnings from 2002-2014.

The large difference in the lower left tail of the earnings distribution between the PSID and LEHD has potential implications for estimated volatility. Workers at the bottom of the earnings distribution are precisely the set of workers most likely to have year-to-year differences in earnings. To test this hypothesis, we construct the share of the total variance of the arc percentage change attributable to each earnings bin vs_b where l_b and u_b are the lower and upper boundaries of bin b.

$$vs_b = \frac{\sum_{\{i,t|l_b < e_{it} \leq u_b\}} (a_{it} - \mu)^2}{\sum_{i,t} (a_{it} - \mu)^2}. \tag{8}$$

Figure 18 - LEHD, PSID, and LEHD-ACS Share of the Variance of the Arc Percent Change in Earnings by Average Earnings Bin (Pooled 2002 to 2014)



Notes: Real annual earnings (2010 PCE) at all jobs for males age 25-59. Each worker must have reported earnings in both the initial and the next available year (subsequent year for LEHD and LEHD-ACS, initial year+2 for PSID). Composite LEHD and LEHD-ACS sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014. The LEHD-ACS sample consists of LEHD annual earnings for all in sample LEHD workers that appear on the ACS during 2002-2014. The PSID sample includes biennial reports of annual wage and salary earnings from 2002-2014. LEHD ventiles were used to create 20 bins. The same earnings bins were used for each data source. The share of the total variance attributable to each bin is plotted at the bin center.

Figure 18 plots the share of the total variance at the center of each bin using a logarithmic scale (base 2) on the x-axis. The LEHD and LEHD-ACS shares are virtually the same, however the variance shares for the PSID differ substantially. A much smaller share of the variance for the PSID comes from workers with earnings below $25,0000, while a larger share comes from workers with earnings above $25,000. Although the results in Figure 18 align well with the earnings distribution results in Figure 17, the differences in the share of variance depend on both the number of workers in the bin and the earnings differences in that bin. To make this clear, we perform a decomposition. First, define the mean squared arc percentage change deviation for bin b as

$$mse(APCS_b) = \frac{\sum_{\{i,t|l_b < e_{it} \leq u_b\}} (a_{it} - \mu)^2}{N_b}, \tag{9}$$

where N_b is the number of persons in bin N_b. Using the mean squared arc percentage change deviation and the share of workers in each bin, vs_b can be decomposed as follows

$$vs_b = \frac{N_b}{N}\frac{mse(APCS_b)}{mse(APCS)}. \quad (10)$$

The share of the total variance in bin b is the product of the share of workers and the ratio of the mean squared arc percentage change deviation in bin b to the overall mean squared arc percentage change deviation. Using this formula, Figure 18 shows the results of an experiment with the PSID data. We set the share of workers in each bin equal to the LEHD values and plot the implied variance shares using the ratio of mean squared arc percentage change deviations from the PSID. Setting the PSID earnings distribution equal to the LEHD earnings distribution yields a simulated PSID share of the total variance for workers with earnings below $25,000 substantially larger than a similar share of the total variance for LEHD workers. Although the PSID has a relatively small proportion of workers at the bottom of the earnings distribution, these workers have a comparatively large relative mean squared arc percentage change deviation compared with workers in the LEHD data.

h. PSID Comparisons

The PSID has been the workhorse data set in the earnings volatility literature and there have been many papers using it to estimate earnings volatility trends (Moffitt and Zhang, 2018a and 2018b; Moffitt and Zhang, 2020). A more in-depth comparison of the impact of differences in cross-sectional earnings distributions in the PSID and the LEHD on estimated volatility trends is therefore warranted. In the previous section, we highlighted the important role workers with low earnings play when estimating earnings volatility. On average, for the LEHD data, workers with earnings less than $7,000 contribute almost 30% of the share of the total variance of the arc percentage change during 2002-2014 compared with less than 15% on the PSID. Given the large differences between the PSID and LEHD data in the share of workers at the bottom of the earnings distribution, estimated volatility differs between the two data sources to a large extent because of the difference in the share of workers, although as we show in Figure 18 average volatility is typically higher in the PSID reducing to some extent the effect of the differences in

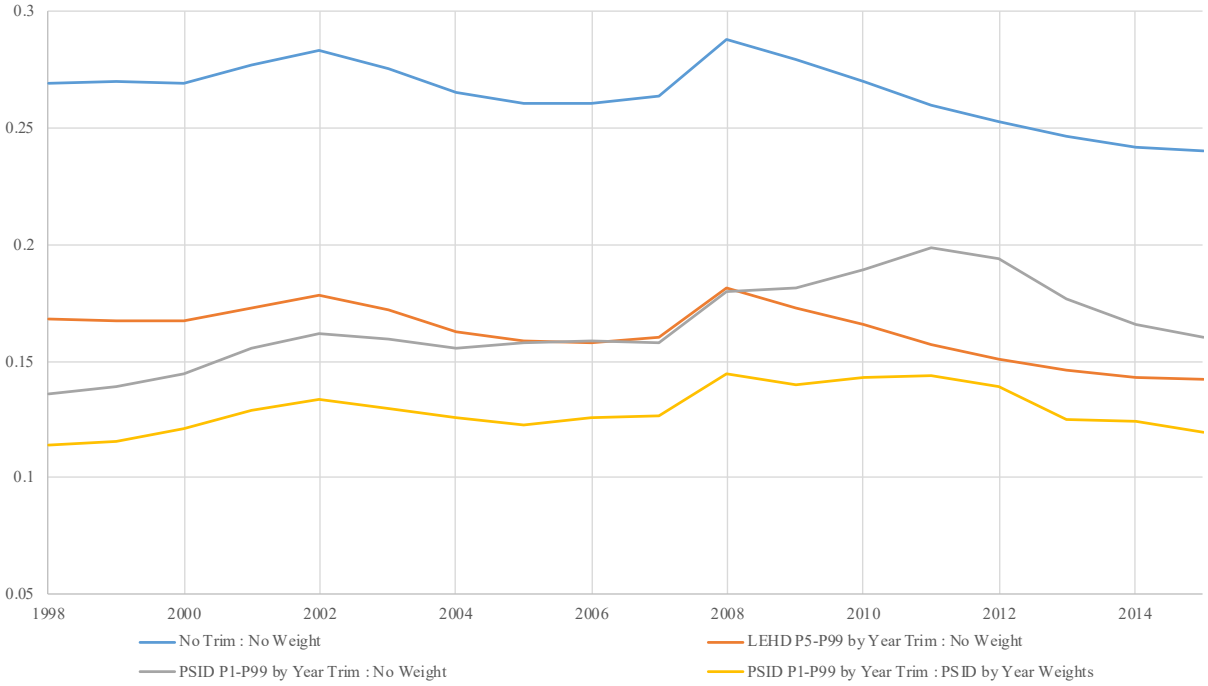
the share of workers with low earnings. Different trends in the share of workers at the bottom of the earnings distribution and different trends in volatility for these workers will both affect the trend in overall earnings volatility. As a complement to the results presented in the previous section and to identify the major factors contributing to differences in the volatility trends between the LEHD and PSID, we calculate LEHD earnings volatility measures using trim values and weights calculated from the PSID.[16]

For the set of LEHD workers with earnings in both years of the year-pair, we create two new earnings trims. Starting with the LEHD data trimmed at the by-year 1st and 99th by year percentiles of the LEHD earnings distribution, we then trim LEHD initial and subsequent year earnings at the PSID by-year 1st and 99th percentile annual real earnings values. Next, we trim LEHD average annual real earnings for the year-pair at either the PSID by-year 1st and 99th percentile average annual real earnings values or the PSID year-2000 average annual real earnings 1st and 99th percentile values. Using the PSID ventiles, we also create two sets of weights designed such that the weighted LEHD annual earnings distribution is approximately equal to either the corresponding PSID average annual earnings distribution or the PSID year-2000 average earnings distribution. [17]

[16] The LEHD comparisons with the PSID shown in Figures 19 and 20 are part of a common effort to assess the impact of the bottom of the earnings distribution on earnings volatility. Each paper in this collection, using a different primary data source, presents similar figures using identical PSID trims and weights from Moffitt and Zhang 2020.

[17] Please see Appendix B for a more detailed explanation of our experiment, including graphs of the trim values and their impact on the sample of LEHD workers.

Figure 19 - Variance of the Arc Percent Change in LEHD Earnings (LEHD and PSID Trims : PSID Weights) by Year

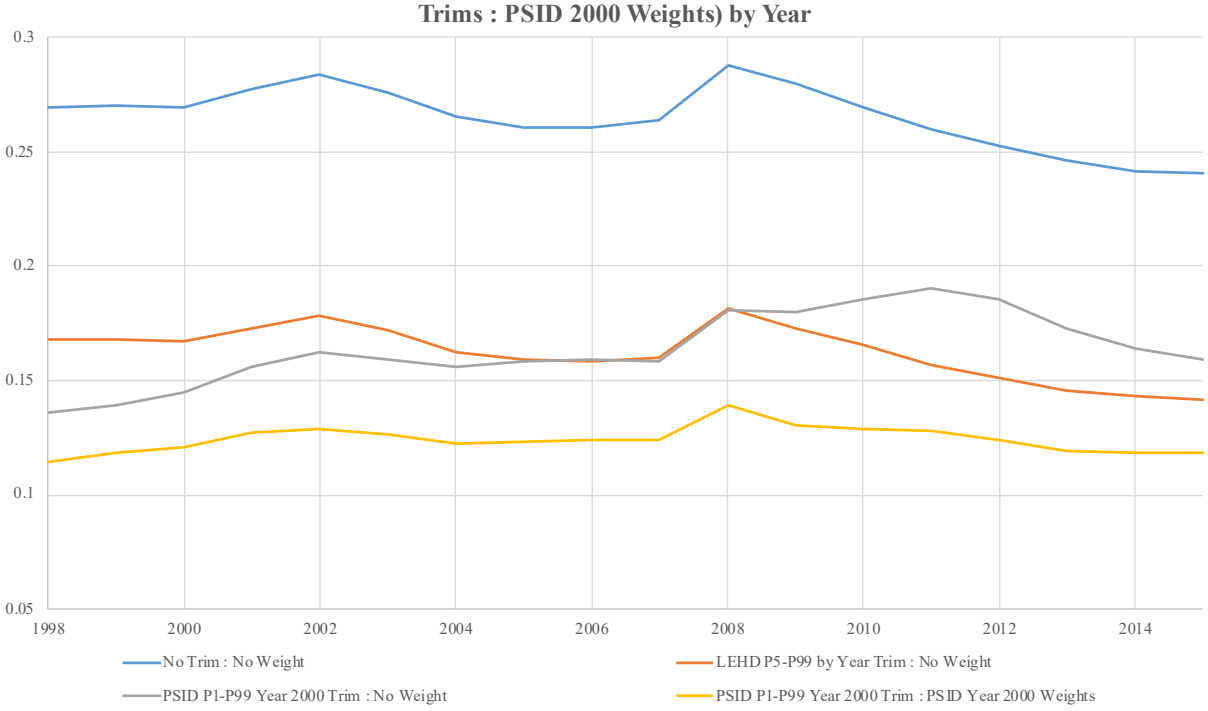


Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. LEHD trim P1 ~ $300 , trim P5 ~ $2,800, and trim P99 ~ $300K. PSID Trim P1 ~ $2,900 and trim P99 ~ $330K. LEHD trim values are stable over time, but PSID P1 trim values decline significantly from ~ $5,000 in 1998 to ~$1,000 in 2014. PSID trims are done separately on year 1 earnings, year 2 earnings, and average earnings. The variance weights are calculated such that the weighted LEHD annual average earnings distributions approximately equal the corresponding PSID annual average earnings distribution.

Figure 19 shows the results of applying the various combinations of PSID trims and PSID weights to the LEHD data. All the series shown in Figure 19 use LEHD earnings data, the PSID data are only used to determine weights and trim values. As a reference point, we plot the no weight, no trim variance of the arc percentage change series shown in Figure 3. Applying just the PSID trim (label: PSID P1-P99 by Year Trim : No Weight), the overall level of earnings volatility falls and the trend in volatility changes from decreasing to increasing. Applying both the PSID trim and the PSID weights (label: PSID P1-P99 by Year Trim : PSID by Year Weights) further reduces the level of volatility relative to the PSID-trim-only series, while also moderating the increasing trend. This is the method used by Moffitt and Zhang (2020) and shows their results should be expected to have a slightly upward volatility trend compared to the LEHD data. For comparison, we also show a series without PSID weights where the initial and subsequent earnings values are trimmed at the LEHD 5th and 99th percentile by year earnings values (label: LEHD P5-P99 by Year Trim : No Weight). Figure 20 shows results from a similar exercise,

instead holding the PSID average earnings trim values and the PSID weights constant at their year 2000 values. The trend is still positive for the PSID trim only series, although somewhat less than in Figure 19.

Figure 20 - Variance of the Arc Percent Change in LEHD Earnings (LEHD and PSID 2000 Trims : PSID 2000 Weights) by Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. LEHD trim P1 ~ $300 , trim P5 ~ $2,800, and trim P99 ~ $300K. PSID Trim P1 ~ $2,900 and trim P99 ~ $330K. LEHD trim values are stable over time, but PSID P1 trim values decline significantly from ~ $5,000 in 1998 to ~$1,000 in 2014. PSID trims are done by year on earnings in year 1 and 2, but the average earnings trim is fixed at the year 2000 values. The variance weights are calculated such that the weighted LEHD annual average earnings distributions approximately equal the year 2000 PSID annual average earnings distribution.

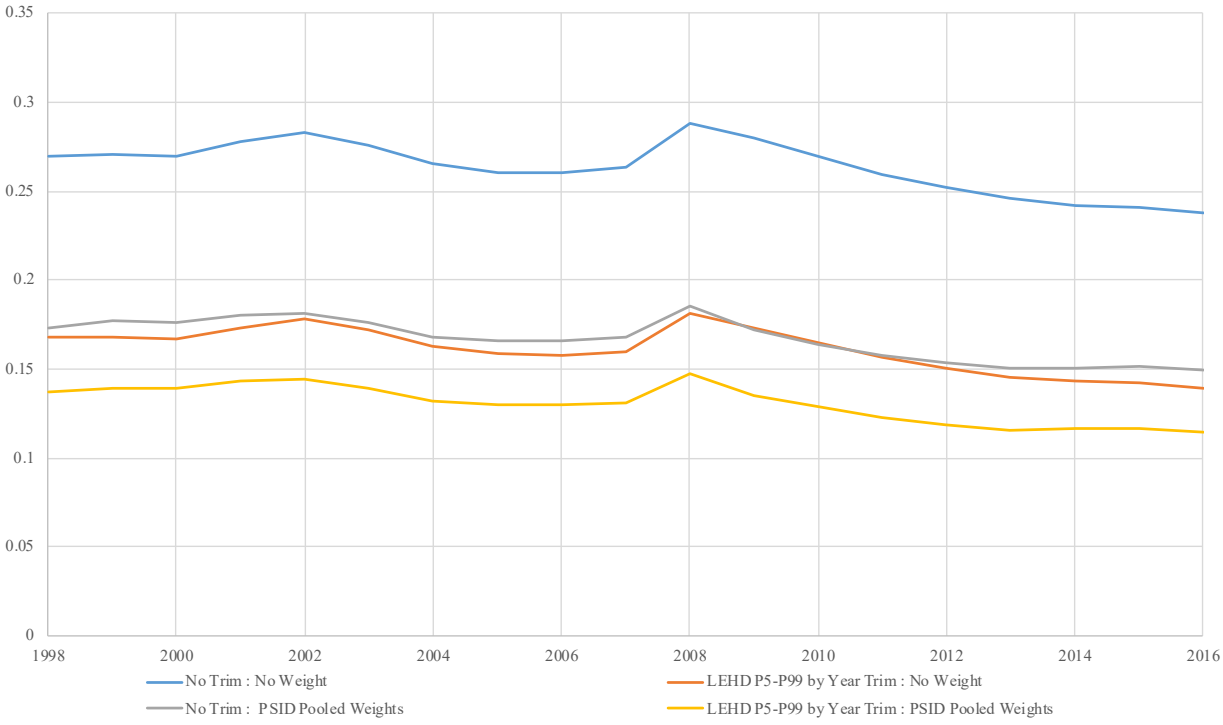
Imposing the PSID trims (label: PSID P1-P99 by Year Trim : No Weight) as shown in Figures 19 and 20 has a relatively large effect on the LEHD trend in earnings volatility (label: no Trim : No Weight), changing it from negative to positive. Unlike in the LEHD data where the by-year trim values are stable, the 1st percentile by-year trim values for the PSID decrease significantly over the analysis period and it is this feature of the PSID distribution that generates the difference. The decrease in the PSID 1st percentile of earnings is large, declining from around $5,000 in 1998 to around $1,000 in 2014, covering much of the region of the LEHD earnings distribution we have already shown responsible for about 30% of the LEHD variance of the arc percentage change in earnings. The LEHD data, on the other hand, shows no decline in the 1st

percentile. The systematic decrease in the PSID by-year trim values increases the estimated LEHD volatility as more low earning workers enter the LEHD analysis sample each year, thus biasing the trend upward. It is important to remember that the trim values sourced from the PSID are based on very small samples and are thus relatively fragile. The PSID per-year sample size is approximately 2,000 workers, resulting in 1st percentile earnings values constructed from about 20 observations. A small change in either the PSID re-contact procedures and/or the earnings questions could have a noticeable effect on the tails of the estimated earnings distribution.

A viable alternative to using the declining per-year PSID 1st percentile trim values should be to apply the year-2000 PSID average earnings trim values to each year of the LEHD average earnings data, as shown in Figure 20, although puzzlingly using constant average earnings trim values does little to change the results. This is primarily due to the construction of the experiment because we first apply the by-year PSID trim to each year of the LEHD data and then apply the year-2000 PSID average earnings trim. The by-year PSID trim has a significant impact on the resulting LEHD sample due to the relatively large number of LEHD workers with average real annual earnings above the year-2000 PSID average earnings trim values, but below at least one of the by-year PSID real annual earnings trim values. The by-year initial and subsequent year trim becomes less restrictive over time as the PSID 1st percentile values decline, removing fewer workers from the bottom tail of the sample each year, increasing earnings volatility relative to what it would be if only the year-2000 PSID average earnings trim was used.

Taking into account the fragility of the annual PSID 1st percentile estimates, we implement an alternative, hybrid approach. First, we create constant PSID-based earnings bin weights using pooled PSID data from 2002-2014 constructed such that the pooled weighted LEHD earnings distribution equals the pooled PSID earnings distribution. The new weights impose the PSID earnings distribution on the LEHD data, but in this case instead of relying on a single year, we pool the PSID real annual earnings data across a large number of years. Second, we combine this re-weighting with an LEHD by-year 5th percentile annual earnings trim and eliminate the secondary average earnings trim used in Figures 19 and 20. The noisy by-year trim based on the PSID 1st percentile is replaced with the more reliable 5th percentile of the LEHD earnings distribution. The 5th percentile of the LEHD distribution of approximately $2,800 is a reasonable proxy for the PSID 1st percentile, lying in the middle of the by-year PSID 1st percentile trim values ($5,000 in 1998 to $1,000 in 2014).

Figure 21 - Variance of the Arc Percent Change in LEHD Earnings (LEHD Trim : PSID Pooled Weights) by Year



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. LEHD trim P5 ~ $2,800, and trim P99 ~ $300K. The variance weights are calculated such that the weighted LEHD annual earnings distributions approximately equal the pooled PSID 2002-2014 annual earnings distribution.

Figure 21 shows the results of the hybrid approach. Unlike the results shown in Figures 19 and 20, neither of the series in Figure 21 show an upward trend in LEHD earnings volatility. Although the levels change depending on the particular combination of trim and weights, either removing low earnings workers (label: LEHD P5-P99 by Year Trim : No Weight) or down weighting them in the variance calculation (label: No Trim : PSID Pooled Weights) results in similar trends. The results in Figures 19-21 reinforce the sensitivity of our volatility measures to the measured left tail of the earnings distribution. Caution should be used when interpreting results from data sources with sparse coverage of low-earning workers.

Figure 22 - PSID and LEHD Variance of the Arc Percent Change by Year



Notes: Real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the current and the next available year (subsequent year for LEHD, current year+2 for PSID). The PSID sample includes biennial reported annual wage and salary earnings from 1998-2014. LEHD composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. LEHD trim P5 ~ $2,800, and trim P99 ~ $300K. PSID Trim P1 ~ $2,900 and trim P99 ~ $330K. LEHD trim values are stable, but PSID P1 by year trim values decline from ~ $5,000 in 1998 to ~$1,000 in 2014.

In Figure 22 we compare PSID earnings volatility with LEHD earnings volatility. This is the only figure in this paper where we calculate the variance of the arc percentage change using PSID data. The results are not as different as one might expect. Using the by-year trim values, volatility trends are positive, while when using a same-each-year trim they are essentially flat. The P5-P99 LEHD series using a by-year trim is similar to the PSID P1-P99 same-each-year trim series except there is a larger decline in earnings volatility post 2004 in the LEHD data than is present in the PSID, although the difference is not large.

IV. Conclusions

The lower tail of the earnings distribution plays a critical role in understanding earnings volatility changes over time. Our carefully constructed linkage of the LEHD and ACS data confirm that the LEHD earnings distribution is cross-sectionally representative, but the PSID earnings distribution systematically underestimates the lower tail. Because of this systematic difference, PSID estimates of log earnings volatility are far more sensitive to data treatments like trimming and re-weighting than are the administrative data estimates. Earnings reports from longitudinal surveys like the PSID may not consistently capture earnings for low labor force attachment, low-earning workers. It might be worthwhile to consider modified earnings questions, supplementing approaches that are designed to capture detailed information on the main job during the year with questions that are designed to capture all labor earnings, as in the ACS. However, there may be limits to the information that respondents can reliably provide, as seen in the differences between administrative earnings lower tails and those of large-scale cross-sectional surveys like the ACS. Combining administrative earnings reports with survey data reports seems like a more reasonable approach to exploiting the strengths of both data sources.

Although total earnings volatility is declining, there appear to be two types of workers, ones with stable attachment to their current employer and those whose attachment is not stable. These two groups experience very different earnings volatility levels and trends. Understanding the mobility between these two groups is an important research activity, greatly aided by producing statistics stratified by this definition of stability.

Finally, the reason most of the movement in earnings volatility is driven by the "not stable" group is the importance of weak labor market attachment for this sub-population. Measures like the arc percentage change capture these very low-earnings periods without inflating their statistical contribution or eliminating their contribution as happens with most trimming methods. Particularly when assessing the patterns of earnings volatility surrounding recessions, ignoring the contribution of those with weak labor market attachment for individuals still at risk to become employed may seriously understate the volatility increases.

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Table 1 - State Availability and Sample Regimes

Count	State	Sample Regime 1	Sample Regime 2	Sample Regime 3	First YYYY:Q Available	Last YYYY:Q Available	Pct 2012.1 QCEW Emp
1	Maryland	X	X	X	1985:2	2017:4	1.83%
2	Alaska	X	X		1990:1	2016:2	0.22%
3	Colorado	X	X	X	1990:1	2017:4	1.70%
4	Idaho	X	X	X	1990:1	2017:4	0.45%
5	Illinois	X	X	X	1990:1	2017:4	4.38%
6	Indiana	X	X	X	1990:1	2017:4	2.19%
7	Kansas	X	X	X	1990:1	2017:4	0.98%
8	Louisiana	X	X	X	1990:1	2017:4	1.41%
9	Missouri	X	X		1990:1	2015:4	1.99%
10	Washington	X	X	X	1990:1	2017:4	2.12%
11	Wisconsin	X	X	X	1990:1	2017:4	2.08%
12	North Carolina	X	X	X	1991:1	2017:4	2.92%
13	Oregon	X	X	X	1991:1	2017:4	1.23%
14	Pennsylvania	X	X	X	1991:1	2017:4	4.44%
15	California	X	X	X	1991:3	2017:4	11.37%
16	Arizona	X	X	X	1992:1	2017:4	1.85%
17	Wyoming	X	X	X	1992:1	2017:4	0.19%
18	Florida	X	X	X	1992:4	2017:4	5.78%
19	Montana	X	X	X	1993:1	2017:4	0.31%
20	Georgia	X	X	X	1994:1	2017:4	2.90%
21	South Dakota	X	X		1994:1	2017:1	0.30%
22	Minnesota	X	X	X	1994:3	2017:4	2.05%
23	New York	X	X	X	1995:1	2017:4	6.49%
24	Rhode Island	X	X	X	1995:1	2017:4	0.35%
25	Texas	X	X	X	1995:1	2017:4	8.10%
26	New Mexico	X	X	X	1995:3	2017:4	0.55%
27	Hawaii	X	X	X	1995:4	2017:4	0.44%
28	Connecticut	X	X	X	1996:1	2017:4	1.26%
29	Maine	X	X	X	1996:1	2017:4	0.43%
30	New Jersey	X	X	X	1996:1	2017:4	2.87%
31	Kentucky	X	X	X	1996:4	2017:4	1.32%
32	West Virginia	X	X	X	1997:1	2017:4	0.52%
33	Michigan	X	X	X	1998:1	2017:4	3.04%
34	Nevada	X	X	X	1998:1	2017:4	0.89%
35	North Dakota	X	X	X	1998:1	2017:4	0.31%
36	Ohio	X	X	X	1998:1	2017:4	3.93%
37	South Carolina	X	X	X	1998:1	2017:4	1.35%
38	Tennessee	X	X	X	1998:1	2017:4	2.03%
39	Virginia	X	X	X	1998:1	2017:4	2.65%
40	Delaware		X	X	1998:3	2017:4	0.31%
41	Iowa		X	X	1998:4	2017:4	1.12%
42	Nebraska		X	X	1999:1	2017:4	0.69%
43	Utah		X	X	1999:1	2017:4	0.91%
44	Oklahoma		X	X	2000:1	2017:4	1.11%
45	Vermont		X	X	2000:1	2017:4	0.22%
46	Alabama		X	X	2001:1	2017:4	1.34%
47	Massachusetts		X	X	2002:1	2017:4	2.55%
48	District of Columbia		X	X	2002:2	2017:4	0.43%
49	Arkansas		X	X	2002:3	2017:4	0.86%
50	New Hampshire		X	X	2003:1	2017:4	0.47%
51	Mississippi		X	X	2003:3	2017:4	0.77%
	Share QCEW 2012.1	89.22%	100.00%	97.49%			

Notes: Each row represents a state or DC. States are ordered by the quarter their data first became available in the LEHD infrastrucuture files. All states and DC are available by 2003:3. The first three colums show the states in each sample regime (a sample regime is a set of states with the same sample entry and exit dates). Regime 1 excludes states that enter after the start of the analysis period and runs from 1998:1 to 2015:4. Regime 2 is the complete data period and runs from 2004:1 to 2015:4. Regime 3 excludes states that exit before the end of the analysis period and runs from 2004:1 to 2017:4. The last column shows the proportion of each state as a percentage of national 2012 month 1 BLS QCEW employment. The last row shows the share of national 2012 month 1 BLS QCEW employment in each regime.

Table 2 - Sample Sizes by Regime, Active Status, and Year

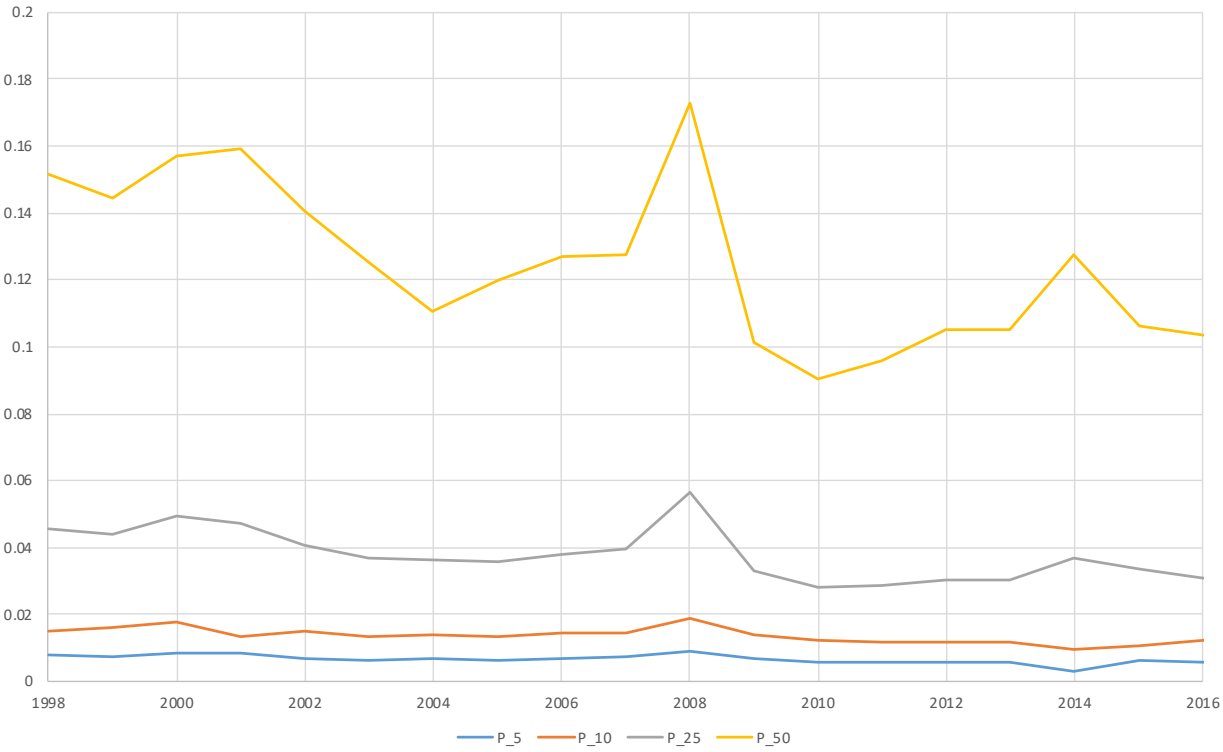
	Active Both Years				Active One Year		
Year Pair	Sample Regime 1	Sample Regime 2	Sample Regime 3	Composite Sample	Exiters-Active Year 1	Entrants-Active Year 2	Total Composite Sample
1998 - 1999	43,180,000			43,180,000	2,874,000	2,818,000	48,872,000
1999 - 2000	43,830,000			43,830,000	2,926,000	2,927,000	49,683,000
2000 - 2001	44,350,000			44,350,000	3,168,000	2,580,000	50,098,000
2001 - 2002	44,040,000			44,040,000	3,574,000	2,463,000	50,077,000
2002 - 2003	43,620,000			43,620,000	3,516,000	2,599,000	49,735,000
2003 - 2004	43,390,000			43,390,000	3,074,000	2,708,000	49,172,000
2004 - 2005	43,560,000	48,860,000	47,570,000	48,860,000	3,340,000	2,919,000	55,119,000
2005 - 2006	43,960,000	49,330,000	48,030,000	49,330,000	3,287,000	2,936,000	55,553,000
2006 - 2007	44,130,000	49,540,000	48,230,000	49,540,000	3,287,000	2,871,000	55,698,000
2007 - 2008	44,130,000	49,560,000	48,250,000	49,560,000	3,503,000	2,667,000	55,730,000
2008 - 2009	43,060,000	48,400,000	47,120,000	48,400,000	4,471,000	2,228,000	55,099,000
2009 - 2010	41,870,000	47,110,000	45,860,000	47,110,000	4,060,000	3,004,000	54,174,000
2010 - 2011	41,960,000	47,240,000	45,980,000	47,240,000	3,431,000	3,338,000	54,009,000
2011 - 2012	42,550,000	47,890,000	46,630,000	47,890,000	3,220,000	3,206,000	54,316,000
2012 - 2013	43,110,000	48,510,000	47,240,000	48,510,000	3,116,000	3,060,000	54,686,000
2013 - 2014	43,680,000	49,140,000	47,850,000	49,140,000	2,985,000	3,019,000	55,144,000
2014 - 2015	44,300,000	49,810,000	48,510,000	49,810,000	2,937,000	2,944,000	55,691,000
2015 - 2016			49,110,000	49,110,000	2,981,000	2,809,000	54,900,000
2016 - 2017			49,630,000	49,630,000	3,021,000	2,855,000	55,506,000

Notes: An observation is a year pair formed from the initial and the subsequent year. To be included, a worker must have reported earnings in the initial and/or the subsequent year. The worker must also be male, age 25-59 in both years, have no more than 12 jobs during the year, appear on the Census Numident, not be reported dead, and have years in the US>0. Counts are rounded to 4 significant figures. Composite LEHD and LEHD-ACS sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016.

Appendix A – Percentiles of the Distribution of the Change in Log Earnings

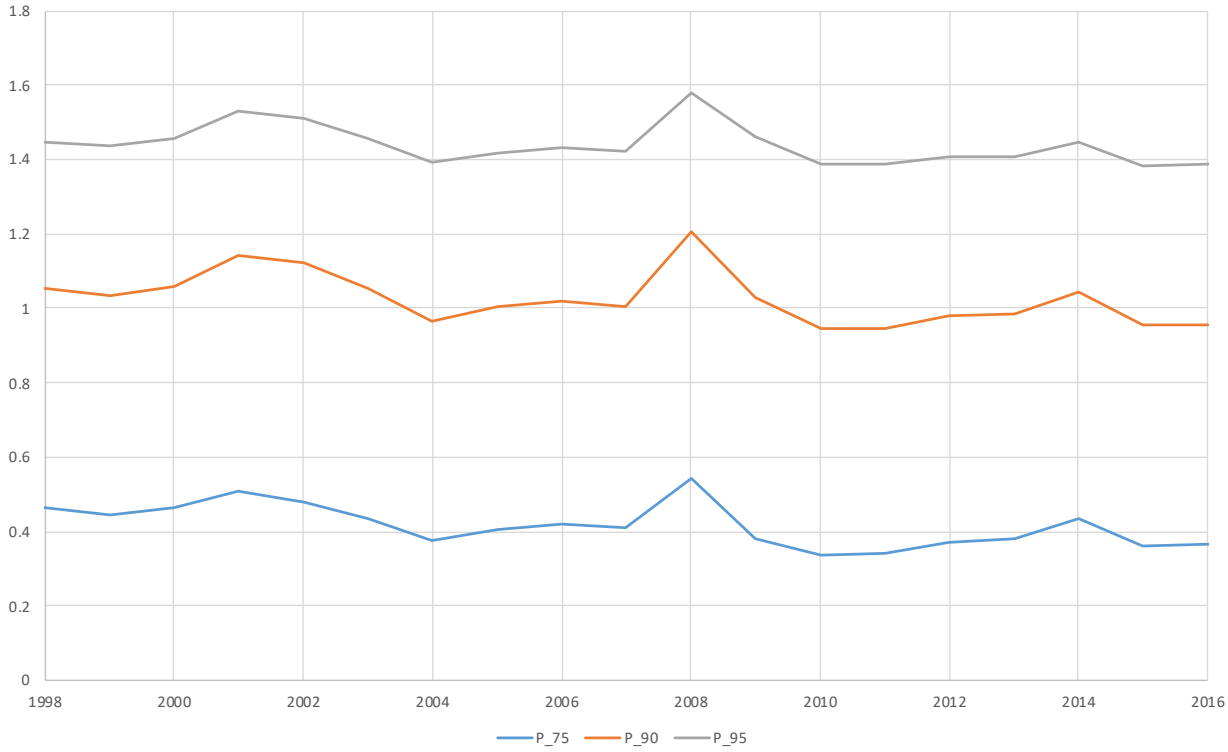
As a supplement to the variance decomposition results presented in section III.e. and the stable worker analysis presented in section III.f. we include the percentiles of the distribution of the change in log earnings $l_{it} = \ln(e_{it+1}) - \ln(e_{it})$ by year. Overall, the percentile trends are relatively stable over time, except for the median (and perhaps also the 25th percentile) which has a general downward trend for both workers with earnings decreases and workers with earnings increases.

Figure A1: Percentiles of the Absolute Value of Earnings Decreases (P5, P10, P25, P50)



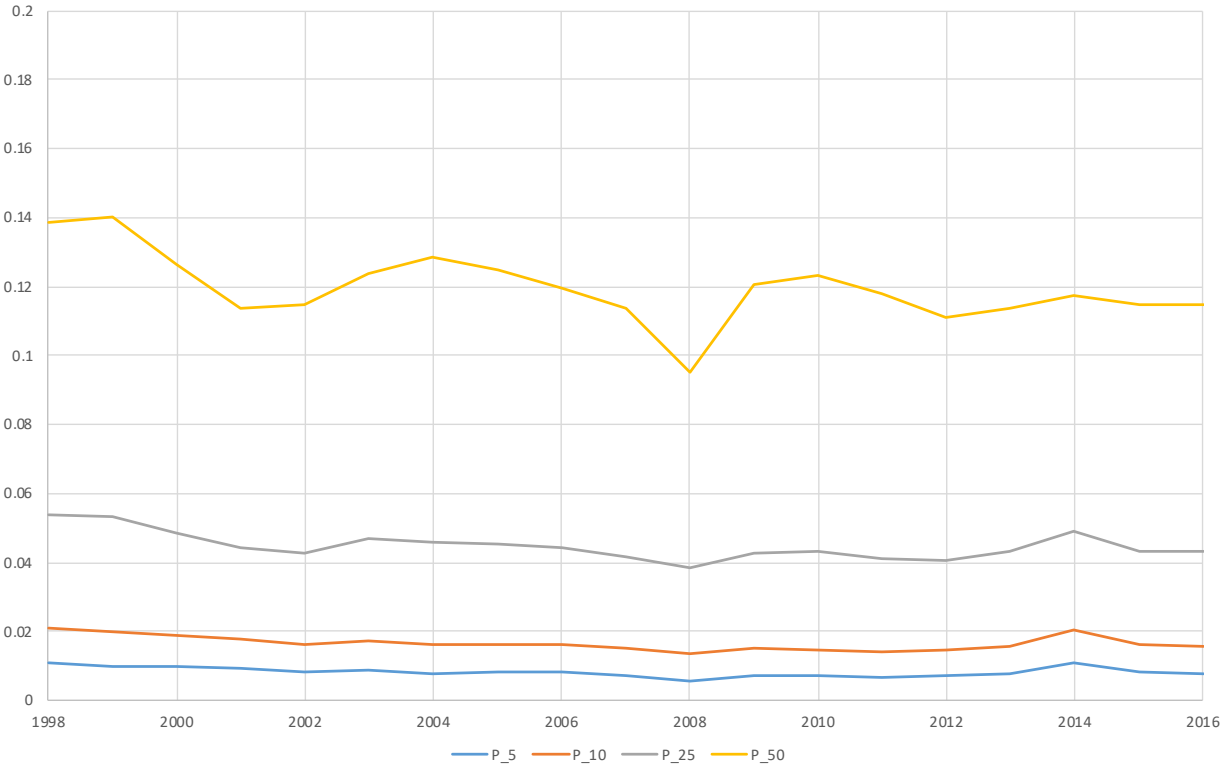
Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. Percentiles are calculated using only workers with a real earnings decrease from the initial to the subsequent year.

Figure A2: Percentiles of the Absolute Value of Earnings Decreases (P75, P90, P95)



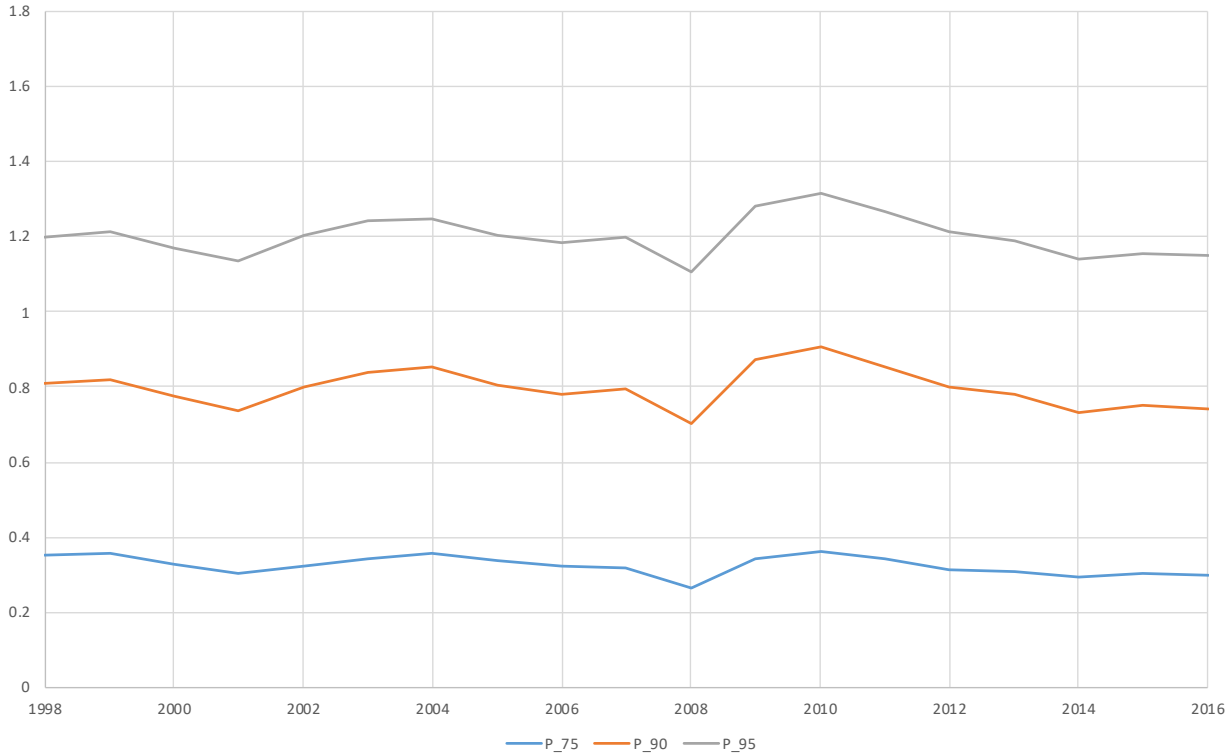
Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. Percentiles are calculated using only workers with a real earnings decrease from the initial to the subsequent year.

Figure A3: Percentiles of the Earnings Increases (P5, P10, P25, P50)



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. Percentiles are calculated using only workers with a real earnings increase from the initial to the subsequent year.

Figure A4: Percentiles of the Earnings Increases (P75, P90, P95)



Notes: LEHD real annual earnings (2010 PCE) at all jobs for males age 25-59. Horizontal axis shows the initial year of the year pair. Each worker must have reported earnings in both the initial and the subsequent year. Composite sample constructed using Regime 1 from 1998-2003, Regime 2 from 2004-2014, and Regime 3 from 2015-2016. Percentiles are calculated using only workers with a real earnings increase from the initial to the subsequent year.

Appendix B – PSID and LEHD Trim Values: Impact on the Analysis Sample

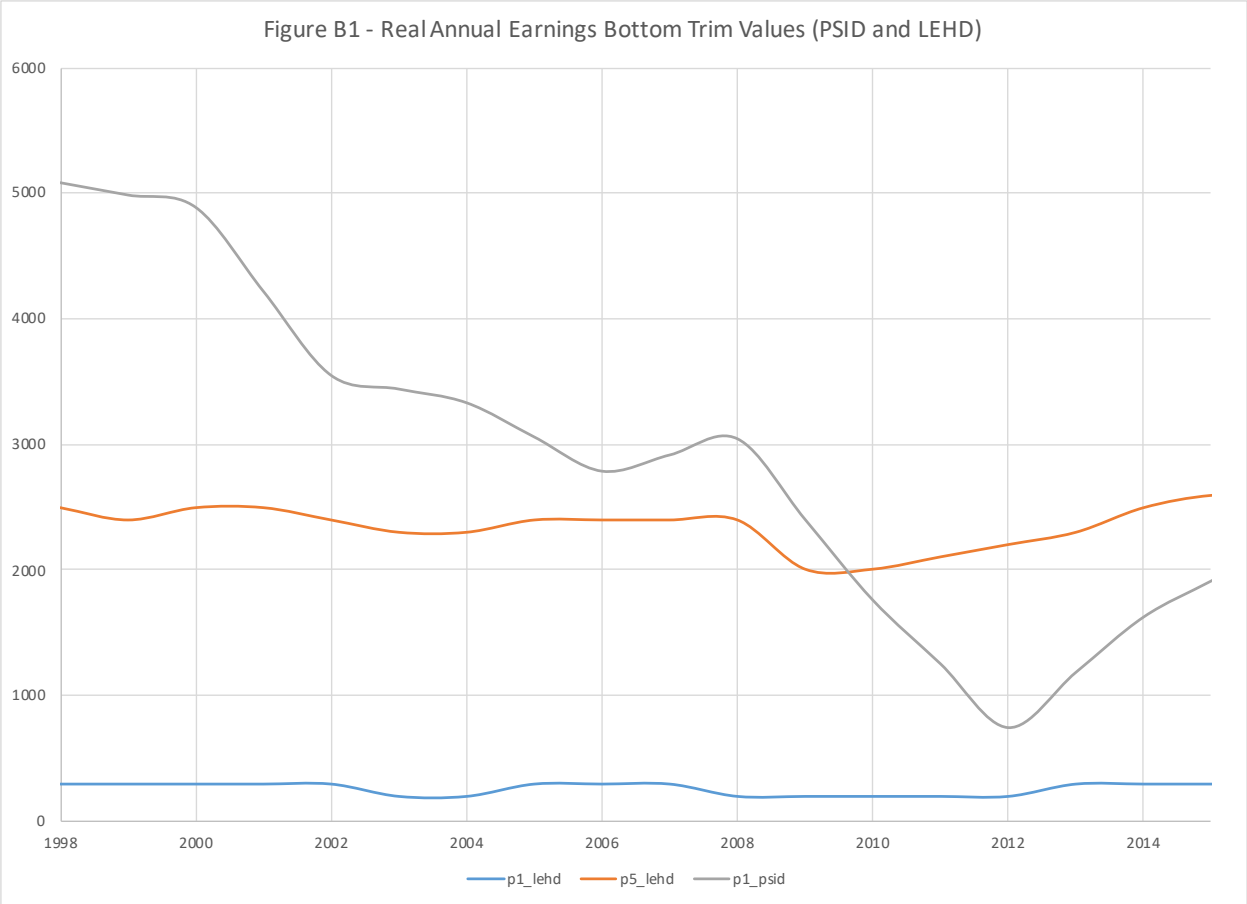
In section III.h. we conduct an experiment designed to adjust our earnings volatility measures for differences in the real annual earnings distribution between the PSID and the LEHD data. The experiment is designed as follows; in the first step, we select a sample of LEHD records comparable to what might be expected to appear on the PSID by applying PSID trimming rules to the LEHD data. In the second step, we use the set of selected records to estimate a weighted earnings volatility measure, where the weights are calculated such that the weighted LEHD earnings distribution matches the unweighted PSID earnings distribution.[18]

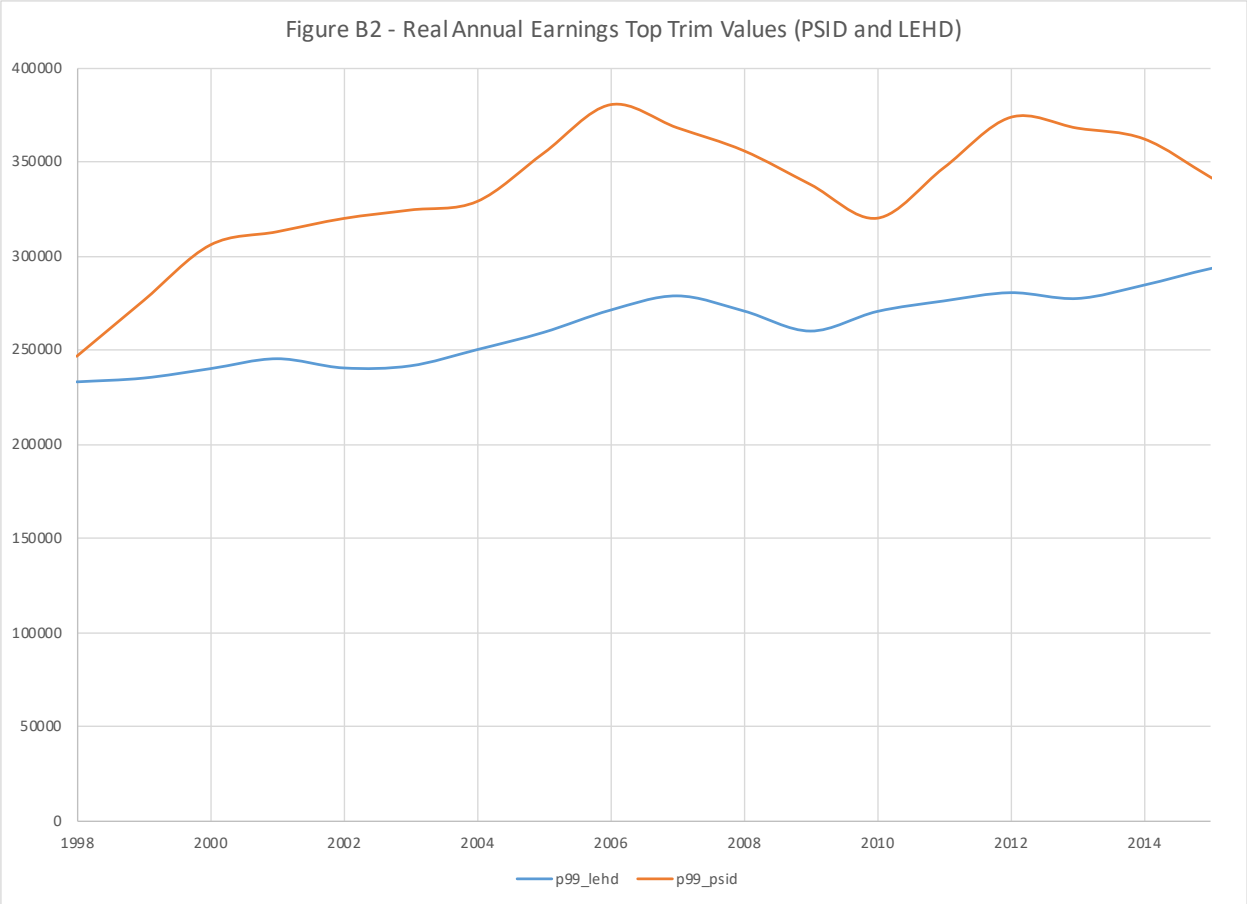
Given the large difference between the left tail of the earnings distribution in the PSID and the left tail of the LEHD data (see Figure 17) and the dominant share of earnings volatility contributed by low earning workers it is worthwhile explaining exactly how the trimming was implemented. The trimming is implemented in multiple stages, with each trimmed value used as an input to the next layer of trimming. As a first step, LEHD real annual earnings are trimmed using the LEHD by-year 1st and 99th percentiles. Next, the initial year and subsequent year LEHD real annual earnings are trimmed using the by-year PSID 1st and 99th initial and subsequent year percentiles. In the final trim, the average of initial and subsequent year LEHD earnings are trimmed at the min/max values of the PSID average real annual earnings distribution (either by year or using year 2000 values for all years). The trims are designed to generate an LEHD sample with the same range of earnings as used in the PSID.

The weights are constructed using either PSID average real annual earnings ventiles or PSID year 2000 ventiles. The weights reflect the relative difference in the proportion of records for each data source in each ventile based bin. For bins where the LEHD data has a relatively large share of earning records, the weight is less than one and is greater than one for bins where the PSID has a relatively large share of earnings records. Figure 17 provides a rough picture of the weights. The weights are typically less than one below around $30,000 and greater than one above that amount (except for the extreme right tail where the weights are about one).

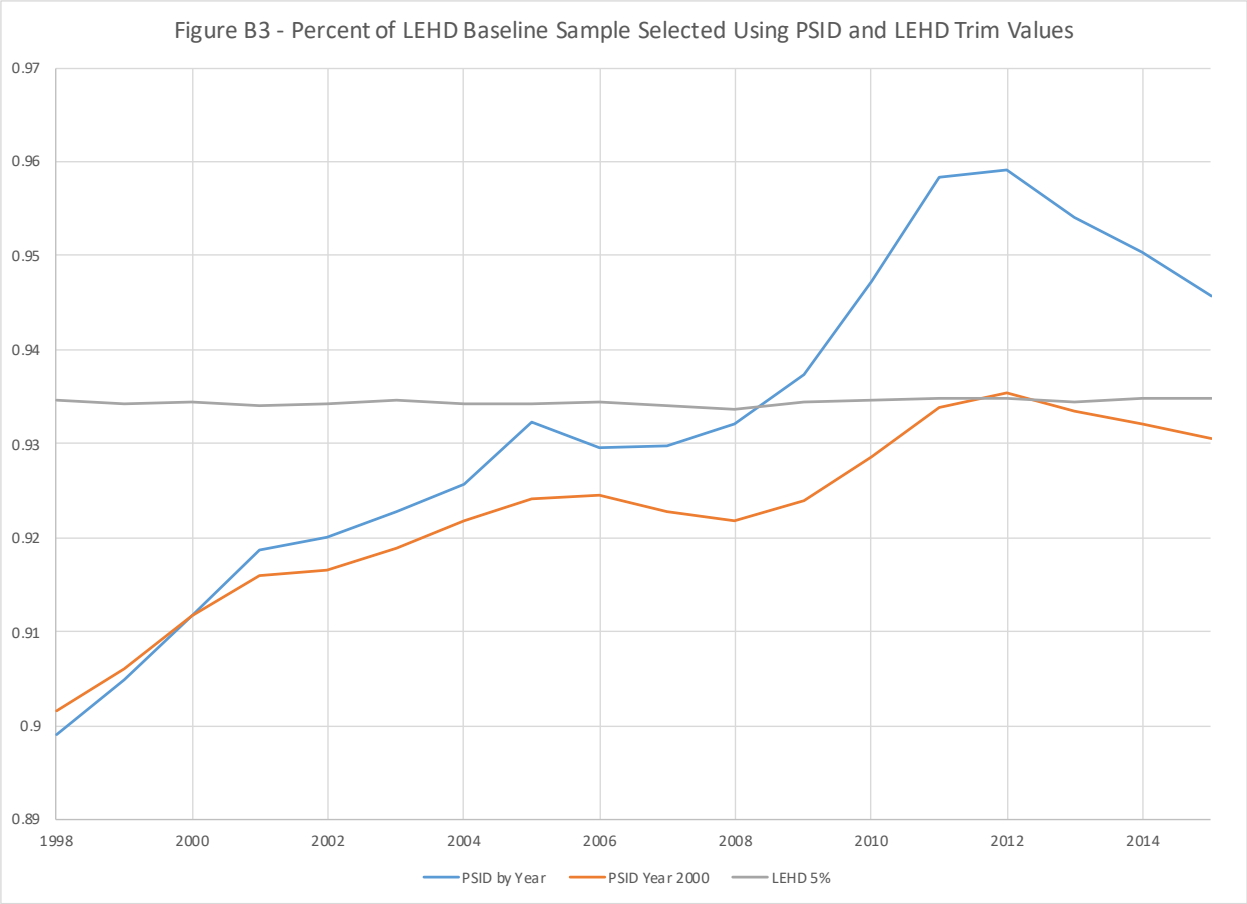
The first layer of trimming using LEHD based trimming values removes approximately the as designed 2% of real annual earnings records each year, while the second and third layers of trimming have a more substantial effect that changes over time. The second layer of trimming has a significant impact due to the declining over time number of LEHD workers with average real annual earnings above the PSID min/max values, but below at least one of the per year second layer PSID real annual earnings trim values. In Figures B1 and B2 we show the second layer PSID bottom and top trim values. In Figure B1 we also include the approximate 1st and 5th percentile LEHD and in Figure B2 we also include the LEHD 99th percentile values for comparison. What stands out immediately is the strong downward trend in the PSID 1st percentile annual earning trim values from 1998 to 2012 compared with the LEHD percentiles which are relatively stable over the period. Interestingly, the 99th percentile trim values have a similar trend between the two data sources.

[18] The PSID trim values and the ventiles used to construct the weights are from Moffitt and Zhang (2020).


Figure B1 - Real Annual Earnings Bottom Trim Values (PSID and LEHD)
6000
5000
4000
3000
2000
1000
0
1998
2000
2002
2004
2006
2008
2010
2012
2014
p1_lehd
p5_lehd
p1_psid



The large difference in the bottom trim values results in a similarly large difference in the percent of the LEHD baseline sample selected each year. In Figure B3 we plot the percent of the LEHD baseline sample selected as the result of the three layers of trimming. Using the by-year PSID trim values, the proportion of the baseline sample selected increases each year up to 2012 due to the inverse trend in the PSID bottom trim values shown in Figure B1. We can also see the positive trend remains even when using the PSID year 2000 third layer trim values. In contrast, applying a 5% LEHD trim on initial and subsequent year earnings instead of the second and third layer PSID based trims, results in a relatively constant percent of the baseline sample selected each year. The overall effect of the combined three layers of trims on estimated earnings volatility changes the downward trend in earnings inequality found in the baseline sample to an increasing earnings volatility trend (see Figure 19), simply by including a larger fraction of low earning high volatility workers in the sample each year from the beginning of the analysis period to 2012.


Figure B3 - Percent of LEHD Baseline Sample Selected Using PSID and LEHD Trim Values
0.97
0.96
0.95
0.94
0.93
0.92
0.91
0.9
0.89
1998
2000
2002
2004
2006
2008
2010
2012
2014
PSID by Year
PSID Year 2000
LEHD 5%